# When Does Admission Control Pay? The Memory Tier Sets the Cost of Concurrency in Agentic LLM Serving

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***Abstract*—Agentic LLM sessions are long, multi-round jobs whose KV footprint grows every round, and on a memory-constrained accelerator the pool is the binding resource. Admission control therefore looks like the obvious lever: a controller that reads engine telemetry and admits on measured capacity should beat a static cap. We give a measurement that says how much room there is, and show what happens when you build against it. Throughput during busy time is $\overline{R}/\tau$—running requests over per-step cost—so raising concurrency by a factor $f$ buys $f^{1-\epsilon}$, where $\epsilon = \mathrm{d}\ln\tau/\mathrm{d}\ln\overline{R}$. Near zero, busy-time throughput keeps converting; near one it does not. It is a property of the workload *and* the engine configuration, not of the controller, and what sets it is the memory tier. The trace bounds reuse at 75.5% of its prompt tokens, and no configuration we ran reaches that bound—the best realizes 92.6% of it. What the tier decides is how much of the bound survives added concurrency: device-only, realization falls from 92.6% to 36.3% as the running count rises from 1.00 to 1.84 and the shortfall appears as recomputation (8.6 M to 20.7 M tokens); with a host tier it holds at 90.2–92.3% between 1.25 and 1.73, while the *device* cache misses more. Two configurations at the same running count and the same pool occupancy realize 78.9% and 91.1% of the bound, so it is not how full the pool is but what happens when it overflows—though that pair is two runs on two days, so we also have a capacity control in the queue that changes the pool size alone on one card; we report it when it lands. Our controllers run on the host-tier configuration, where $\epsilon$ stays below one across their whole operating range, so the ceiling we measure on device-only caps is not what bounded them. It bounds what added concurrency is worth, not what a whole run achieves—idle time and the SLO axis are separate. We measure $\epsilon$ on a memory-bound agentic replay (200 sessions, one 24 GB GPU, a pool that holds four median sessions) and get $0.38$ to $0.61$ depending on which step cost a system can observe: every 10% of added concurrency buys 4–6% of throughput.**

**The verdict on the controllers is negative, and we report it at that strength. (The criterion of §IV is a diagnostic and does not return this verdict; §IV is explicit that it does not explain the failure.) Nine controller variants spanning four routes, plus a reimplementation of an independent congestion-based controller, none is above the static Pareto front at the latency threshold interactive serving cares about—the best misses it by under a point, on the below side of our tie band, and its repeat misses by 5.6; a plain cap on resident sessions remains the fastest policy we measured. That is not an argument for admitting everything: with no control at all the median request waits 1,774 seconds for its first token, the prefix cache's realized hit rate collapses from the trace's planned 75.5% to 1.6%, and SLO falls to 3%. The question is which selective rule, not whether to be selective. Two measurable effects, rather than one we can tune away, account for part of why our variants fail—the arm-by-arm attribution of §V-C attributes three of the twelve controller arms we tested, leaves six with a step cost that no matched comparison can confirm, and leaves three as residuals we can measure and cannot yet explain. The single step-cost attribution it does make is on a run flagged for host co-tenancy but not measurably damaged by it, and we say which is which rather than counting it. They hold nearly half the pool idle while sessions queue—their own headroom estimate on the base configuration is 1.79× the truth (1.24–2.00× across the controller arms on this trace, and 1.01–4.11× over every controller run we logged, across all three trace families)—and the per-step elasticity measured on the configuration they actually run stays below one across their whole operating range, so the engine had room they did not take: the ceiling we measure on device-only caps does not survive on the host-tier configuration the controllers use. The wall clock splits into busy and idle time; admission policy moves only the running count, and $\tau$ rises with it. The idle component (7.4–36.3% of the sampled span) comes from gaps the tool waits set—an agent session issues no request while a tool runs, and that wait averages 16.85 s measured over the controller's run (the trace demands 15.82 s; they differ by the controller's own admission delay), not the sub-second figure the trace's median suggests—though how many such gaps a run collects is partly the policy's, since a policy sets how many sessions stay live. On device-only configurations we can account for the rise token for token; on host-tier ones part of it is co-tenancy we found late and part we still cannot attribute, and we say which is which. We also report the six of our own claims we retracted along the way, and the boundary of the verdict: at a loose latency threshold one variant does cross the front, by a margin we cannot resolve against run-to-run noise.**

***Index Terms*—LLM serving, admission control, KV cache, agentic workloads, performance characterization, negative results**

## I. Introduction

An agentic LLM session is not a request. It is a long-lived job that issues a round of inference, waits on a tool, and comes back—tens of times—with a context that grows every round. On a memory-constrained accelerator this makes admission control look like the obvious lever. The KV pool is the binding resource; a single large session can occupy most of it; and the sessions themselves are wildly heterogeneous. If a scheduler can see how much of the pool a candidate will need and how much room is left, surely it should beat a fixed number.

We believed this, built a controller that does exactly that, and spent a considerable amount of GPU time discovering that on our workload and hardware it is false. This paper reports that negative result, and the negative result is most of what we have. Alongside it we report one measurement—the elasticity of per-step cost with respect to the running count—that we hoped would say in advance how much room admission

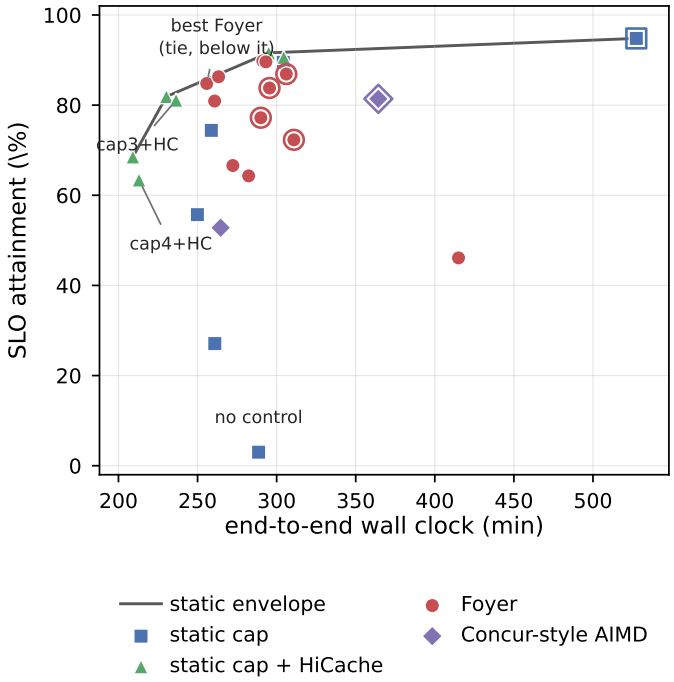



Fig. 1. Every completed run on the (wall clock, SLO) plane. The grey line is the upper-left envelope of the static points—drawn from the data, so it uses each configuration's *better* run where a repeat exists, which is why `cap3`+HiCache's 236-minute run sits below the line rather than on it; a policy below or on it has not beaten a static cap, because time-slicing two caps approximates any point on the segment between them. Nine Foyer variants plus a reimplementation of an independent controller, none above at the 10 s latency threshold used throughout; §IX reports how far that verdict is conditional on the threshold. The four points drawn with a hollow ring were measured while three other engine instances shared the host, which inflates per-step cost and makes them incomparable with the rest (§IX-0e); they are plotted for completeness and are being re-measured.

control has. It does bound that room on the configurations we measured, but when we tested whether it explains our controllers' failure, on their own configuration and on a second workload, it did not. We report it as a diagnostic of a workload, not as a predictor of whether a controller will win.

### A. What we measured

We serve a replay of 200 real coding-agent sessions on a single RTX 3090 (Qwen2.5-Coder-7B, 96K context, SGLang, a 101,432-token KV pool) and compare a family of admission policies against a sweep of static concurrency caps. Nine controller variants, spanning four mechanistically distinct ideas, all land below the static Pareto front, and so does a reimplementation of an independent congestion-based controller from the literature (Fig. 1). The best of them misses by under a point—inside the $\pm 1.0$ pp band we call a tie, but on the below side of it—and its repeat misses by 5.6.

### B. Why

We decompose the wall clock into *busy* time, during which the engine has at least one request in flight, and *idle* time, during which it has none. The busy component is governed by $W_{\text{busy}} \approx (G/\overline{R})\,\tau$ (Eq. 3), where $\overline{R}$ is the mean number of running requests, $G$ the generated tokens and $\tau$ the per-step cost. An admission policy moves $\overline{R}$, and on our configurations it moves little else—`bres` is the exception, and §III uses it to show why moving both components at once is still not enough. The difficulty is that $\tau$ is not constant in $\overline{R}$: across our runs $\tau_1$ spans 27.0–36.7 ms, and on the device-only family, raising the busy-time running count from 1.00 to 1.84 absorbs a third of what the extra concurrency buys.

The reason $\tau$ moves is recomputation, and whether recomputation happens is decided by the memory tier rather than by the policy. A session's prefix is worth reusing only while it is still resident, so prefix *reuse* is bounded by the workload—in our trace, 75.5% of the prompt tokens, a bound no configuration we ran reaches. How much of that bound is realized is where the tier enters. With the KV pool on the device alone, realization falls as concurrency rises—92.6% of the bound at $\overline{R}_{\text{busy}}$ 1.00 down to 36.3% at 1.84—and the shortfall appears as recomputation, 8.6 M to 20.7 M tokens. With a host tier in front of it, realization holds across the range our controllers occupy—90.2% to 92.3% between 1.25 and 1.73—even though the *device* misses more. Two configurations at the same running count and the same pool occupancy realize 78.9% and 91.1% of the bound respectively (§III); what separates them is not how full the pool is but what happens when it overflows. That pair also differs in date and in GPU, so a capacity control that changes the pool size alone on one card is queued behind the re-runs; §V reports it when it lands. That is the finding we would carry to another system first, and it is why this paper's answer to its own title is a property of the deployment rather than of the controller.

The idle component is structural rather than a control failure. A session in a tool wait issues no request, so when all live sessions happen to be waiting, the engine is genuinely empty. The stretches are short in the middle and long in the tail—mean 20.6 s—and the tail is what a scheduler feels: 69.6% of all idle *time* sits in stretches over 20 s, an order of magnitude above the 2 s control cycle. (The median is not a good summary here: it moves between 5 and 10 s depending on which runs are pooled, because one long run contributes a fifth of the stretches.) Crucially, the tool wait itself is much longer than the trace's median suggests: the median inter-round gap is 0.35 s, but the *mean* is 15.8–16.9 s with 22% of gaps over ten seconds. A controller that reads only the median concludes, as we initially did, that sessions never leave and therefore that there is no idle time to recover. There is.

### C. Contributions

- **We identify the memory tier as what decides how much of the achievable prefix reuse survives added concurrency, and therefore how much a controller has to gain by admitting more.** The two senses of "room" need separating, and §VI is what separates them: the tier bounds the *local* return on added concurrency (busy-time throughput), and it does not by itself decide whether a controller wins *end to end*, where idle time and the SLO axis also move. Our second workload is

the demonstration—it has no host tier and runs at low concurrency, the two conditions this section associates with room, and its controller still does not win (§VI). The claim here is the local one. The trace bounds reuse at 75.5% of its prompt tokens, and no configuration we ran reaches that bound—the best realizes 92.6% of it. What the tier decides is how much of the bound is realized as concurrency rises: device-only, realization falls from 92.6% to 36.3% between $\overline{R}_{\text{busy}}$ 1.00 and 1.84 and the shortfall appears as recomputation, 8.6 M to 20.7 M tokens; with a host tier it holds at 90.2–92.3% over 1.25 to 1.73, while the *device* misses more (§III). Two configurations at the same running count and the same pool occupancy realize 78.9% and 91.1% of the bound, so it is not how full the pool is but what happens when it overflows; that pair differs in date and GPU as well, so a capacity control that varies the pool alone on one card is queued; §V reports it when it lands. The per-step elasticity follows the same line: it passes one above $\overline{R}_{\text{busy}} \approx 1.5$ device-only and stays below one across the controllers' whole operating range with a host tier (§IV). Two measurements that do not share a code path—client-side step timing and engine-side token accounting—give the same answer. This is the finding we would carry to another system first, and it is what the title's question turns on.

- We test four routes by which an adaptive admission controller was expected to beat a tuned static cap on this workload, and we are precise about what each one rests on. Two are falsified by measurement against criteria we fixed before the runs—changing the accounting base (§V) and changing the growth reserve. One is a knob rather than a hypothesis, and we report it only because it slides the operating point along the front and adds no mechanism. The fourth, changing the execution granularity from session count to running-request count, is falsified by measurement *and* by an argument about why the permit mechanism cannot express the control it would need. We do not call the last two of these falsifications of a stated hypothesis, because they were not.
- We give a two-part decomposition of the wall clock (§III) that locates the cancellation: admission policy moves the running-request count, and the per-step cost moves with it. On the device-only configurations we can account for that rise token for token—evicted prefixes are recomputed almost one for one. On host-tier configurations recomputation is flat and the account has to come from elsewhere; we report the part of that residual we were able to attribute, and the part we were not (§III, §IX-0e).
- We reduce the question to one measurable number, the elasticity of per-step cost with respect to the running count (§IV), and measure it at $0.38$ on the model-consistent step cost and $0.61$ on the per-request one, on the configurations whose mechanism we can account for token for token. We offer it as a diagnostic rather than a predictor: it says where on the curve a policy operates and how much the remaining headroom is worth, and §IV is explicit about the sense in which it is circular and the sense in which it is not.
- We characterize when adaptivity *could* have paid, and show that our workload is not that case: the same throttle configuration that is worth $+51\%$ at 25 sessions costs $-42\%$ at 200 (§VII). Put differently, no single scalar cap is optimal across both, which is the regime adaptive control exists for. On the low-load tier, where that argument bites hardest, the controller is *not* worse—same-generation measurements put it on the front, $+0.9$ pp above the line and inside the noise. On the high-load tier it is dominated. That asymmetry, not a blanket failure, is what this paper measures.
- We make the measurements reproducible: the front and each controller's signed distance are produced by one script from the raw per-step logs, the tables and the sub-thousand-step runs by two others, all named in §IX-0e. We report the six claims we had to retract along the way, with the checks that caught them.

### D. What we do not claim

We do not claim that admission control is useless in general. We claim it does not pay *here*, we show the measurement that decides it, and we argue (§IV) that the same measurement can be taken on another workload before committing to a sweep. We also do not claim our hardware and model are representative; our results are from one 24 GB accelerator and one 7B model, and extending them is future work.

## II. Background and Setup

### A. Workload

We replay 200 coding-agent sessions drawn from the TRACELAB corpus [1]. Each session is a sequence of rounds; a round presents a prompt of median 23,187 tokens (p90 51,064; maximum 89,436, which alone is 88% of the KV pool), and between rounds the session waits on a tool before returning with a longer context. Arrivals are Poisson at $\lambda = 0.04\,\text{s}^{-1}$ over an 80-minute window, giving 999 steps in total. Two sessions of maximum size cannot be co-resident: their combined contexts exceed the pool by 76%.

The tool wait is the load's defining property and the one most easily misread. Its *median* is 0.35 s, which invites the conclusion that sessions never leave the engine. Its *mean* is 16.85 s measured and 15.82 s demanded, and the maximum is 300 s; 21.8% of the tool waits the trace specifies exceed ten seconds, which is 14.4% of all round transitions once the zero-wait ones are counted, or 18.6% on the measured gaps. We verified that our replay harness honors the trace faithfully: across 799 round transitions the measured gap is 16.85 s mean against a demanded 15.82 s. On a typical transition the harness adds nothing—the median of the per-transition measured/demanded ratios is 1.001—while the ratio of the two means is 1.065, the difference sitting in the tail. The corpus's own characterization makes the same point qualitatively—it reports "diverse and heavily-tailed tool calls" and calls these

gaps "human-paced"—but gives no numeric mean [1], so the 16.85 s above is our measurement over the released trace, not a figure we quote from that paper.

### B. Engine and the binding constraint

We use SGLang v0.5.10 [2] with Qwen2.5-Coder-7B-Instruct [3] extended to 96K context, on one RTX 3090 (24 GB). KV is paged [4]; at `mem-fraction-static` 0.85 the pool is 101,432 tokens. Hierarchical caching (HiCache), with a host tier of twice the device pool, is enabled on the configurations marked *+HC* and on every Foyer run; the five plain static caps run without it. We keep both because the host tier turns out to change the mechanism (§III). Every run is guarded: if the pool does not come out at $101{,}432 \pm 20$—which happens when another tenant holds memory on the card—the run is aborted rather than recorded, because SGLang sizes the pool from free memory at startup and a shrunken pool silently produces incomparable numbers.

The relevant consequence is that the pool holds only about four median sessions. Concurrency is therefore capacity-bound, not scheduler-bound, and this is the premise on which everything below rests.

### C. The controller under test

Foyer is an admission controller that reads engine telemetry and issues *named* permits rather than a scalar cap: a session is admitted by appearing in a permit file, and the runner refuses to start a round for a session that is not listed. This design choice matters for the comparison, because it is what allows per-session actions at all—including the pause primitive we exploit in §V-H. It also means that a session, once admitted, holds its permit for its entire lifetime, so the controller never frees capacity by discarding a session's state—it admits fewer sessions instead. That is a statement about the *policy*, not about the memory: the engine's radix cache still evicts parked prefixes under pressure, and we measure that it does and that they are recomputed (§III). We return to this distinction in §IX, because it bounds how far our negative result generalises.

Each control cycle (2 s) the controller computes

$$\Pi \;=\; \max(\rho, \textstyle\sum_i c_i) \;+\; \sum_i g_i, \tag{1}$$

where $\rho$ is the engine's reported occupancy, $c_i$ the context length of live session $i$, and $g_i$ its forecast growth over the next `horizon` rounds. A session is admitted when $\Pi + n \le B$, with $n$ the candidate's estimated need and $B$ the budget. The three terms are respectively the measured physical occupancy, a reservation for sessions currently parked in a tool wait (whose KV is evictable and therefore absent from $\rho$), and a reservation for future growth.

### D. Baselines and the verdict criterion

Baselines are static caps: at most $N$ sessions resident, $N \in \{1, \ldots, 5\}$, with and without HiCache. We compare on two axes: end-to-end wall clock (time from the first submission to the last completion) and SLO attainment (fraction of requests whose time-to-first-token is under 10 s).

The verdict criterion deserves care, because "faster than cap-$N$" is not the question. The question is whether a policy reaches a point *above* the static front. A point that lies merely between two static caps is not a win: mixing cap-$N$ and cap-$M$ by time slice approximates any point on the line between them. We flag this segment argument as an argument, not a measurement—we did not run the time-sliced policies, and carrying queue state and cache contents across a switch could bend the line. It is the criterion we apply, and it is the stricter of the two we could have used—stricter than comparing each controller against the discrete static points alone, because the line sits above them. We state the direction plainly rather than claim it as a caution: this choice makes our negative result *easier* to reach, not harder. Our best arm is 0.93 pp below the line and would be $+3.7$ pp against the nearest static point below it in wall clock (`cap3`+HiCache's first run, 81.1% at 236.4 minutes; $+2.9$ would be the comparison against that anchor's repeat at 81.9%, which is the one we displaced it with). So we take the upper-left envelope of the static points and report each controller's signed distance above or below it. We classify a signed distance of more than $+1.0$ pp as above the front, less than $-1.0$ pp as below it, and the band between the two as *on* it—a tie, not a win. The band is a convention we fixed before running anything, and we report it here because every verdict in this paper depends on it. Our later replicate measurements (§V) bracket it rather than justify it: the SLO spread of configurations we ran more than once ranges from 0.5 pp to 5.1 pp, so for some configurations the band is generous and for others it is far too narrow. We flag this as a weakness of the criterion, not a defence of it. One further property of the envelope deserves flagging, because four of the nine controller variants finish slower than 295 minutes. Its right-hand segment runs from `cap2`+HiCache at 295 minutes to `cap1` at 527, and `cap1` is the only anchor there that runs *without* HiCache. Those four—`hz2`, `bres`, `breshz1` and `foyfix`—are therefore judged against an interpolation between one host-tier point and one device-only point. If the envelope is drawn from the host-tier anchors alone, that segment does not exist and those four cannot be adjudicated at all; drawn from device-only anchors alone, they sit further below. We report the conservative reading and note that the honest statement about those four is *below the front as we defined it*, not *below every static configuration*. Three of the four (`hz2`, `bres`, `breshz1`) were also measured under host co-tenancy and are marked ‡ in Table I; whether they are slower than 295 minutes at all is one of the things the re-runs will settle. The front and every controller's signed distance are computed by one script from the raw per-step logs; the table generator computes the distance for the four runs that did not complete all 999 steps—both are named in §IX-0e—and that script's own first version was wrong: it extrapolated past the end of the envelope and declared the slowest static cap a winner. We fixed it and report the episode because it is the failure mode this criterion is most prone to.

## III. Decomposing the Wall Clock

### A. Two components

We split the wall clock at every sampled instant according to whether the engine has any request in flight:

$$W = W_{\text{busy}} + W_{\text{idle}}. \tag{2}$$

Fig. 2 shows the split for every completed run. Two things are immediately visible and both matter later.

First, idle time is not noise. Across the runs it spans 7.4% to 41.9% of the sampled span—7.4% for the fastest configuration, 36.3% for the slowest, and 41.9% for the one that idles most (all-throttles-off). The denominator here is the span the sampler observed, which runs about six minutes longer than the request wall clock of Table I, so the fractions against that wall clock are 7.6% and 36.7%—and it is not spread evenly. Over 1,849 stretches the mean is 20.6 s and 17.6% of them exceed 20 s, accounting for 69.6% of all idle *time* (Fig. 3). The tail, not the middle, is what a scheduler feels, and it sits at the scale of the 15.8–16.9 s mean tool wait (demanded and measured; `toolwait.csv`). We report the mean and the share rather than the median here because the median is not robust to which runs enter the pool: it is 10.0 s without `cap1` and 5.0–5.6 s with it, since that one run contributes 411 of the 1,849 stretches and is itself the longest. The mean (18.3–21.0 s) and the over-20 s share of idle time (64.8–70.6%) move by little across the four poolings we report in `idle_pooled.csv`, so the claim that the tail dominates does not depend on the choice. The 90th percentile moves with the median (25.1–30.1 s) and we give it as a range for the same reason.

Second, the two components move independently. `bres` (§V) reduces idle from 15.9% to 10.1%, recovering about 17 minutes—and is nonetheless *slower* than its own baseline, because its busy time grows by 21 minutes. Any account that treats "more concurrency" as one thing will get this wrong. We flag one caveat on the numbers in this paragraph: `bres` is one of the runs measured under host co-tenancy (§IX-0e), and co-tenancy inflates $\tau$, which lengthens busy time—so the 21-minute growth is an upper bound on the real one. The qualitative point, that the two components move in opposite directions, is not affected: idle falls by construction under a policy that keeps more sessions live, and the run is slower overall, which is where the comparison starts. The re-run (§IX-0e) will give the corrected split.

### B. What determines busy time

During busy time the engine executes steps; each step produces one token per running request. Hence

$$W_{\text{busy}} \approx \frac{G}{\overline{R}}\,\tau, \tag{3}$$

with $G$ the total generated tokens (fixed by the trace at 706,010) and $\tau$ the per-step cost. Across our runs $\tau$ is not a fit parameter we tuned; we obtain it two independent ways, and reporting both is what shows they are measuring different things rather than confirming each other.

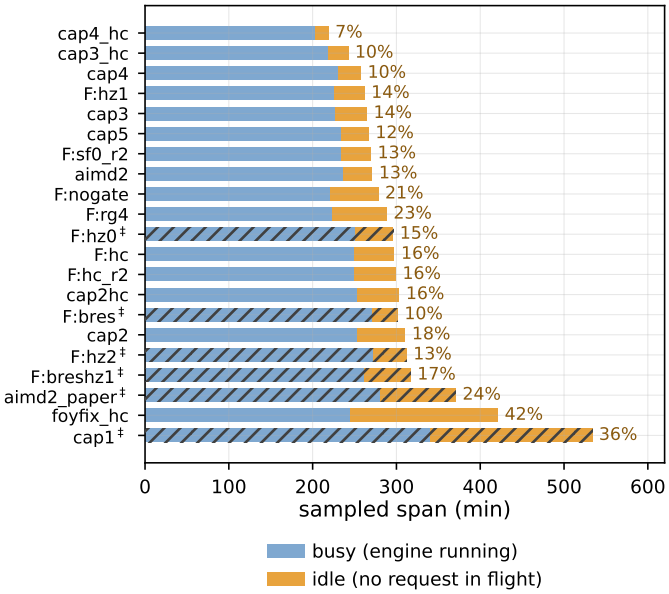



Fig. 2. Wall clock split into busy and idle for every completed run, ordered by total duration. Idle is 7.4% for the fastest configuration and 36.3% for the slowest. Note `bres`: it cuts idle to 10.1% and is still slower than its own baseline, because busy time grows instead—and note that `bres` is one of the six hatched bars, whose busy time is itself inflated by host co-tenancy (§IX-0e); the hatch is the same ‡ flag as in Table I.

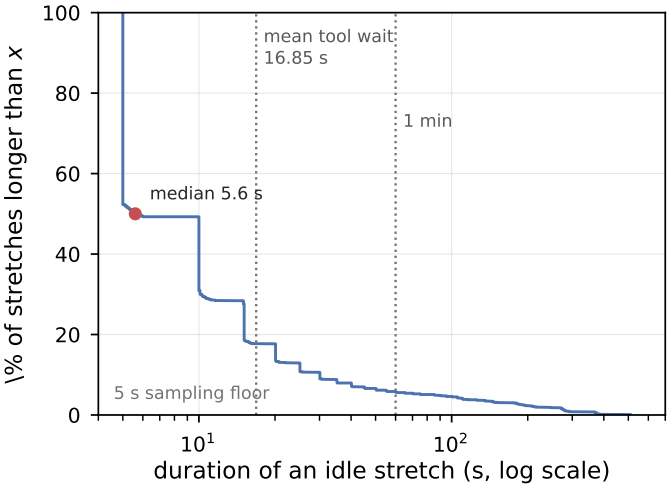



Fig. 3. Survival function of idle-stretch duration, pooled over all completed runs (1,849 stretches). Most stretches are short—the median is 5–10 s depending on which runs enter the pool, which is why we do not lead with it—but the minutes are not: 69.6% of all idle *time* sits in stretches longer than 20 s (64.8–70.6% across the four poolings in `idle_pooled.csv`), and the mean stretch is 20.6 s, the same order as the 15.8–16.9 s mean tool wait. The cliff is the 5 s sampling floor.

*a) Method 1 (derived).:* $\tau_1 = W_{\text{busy}} \cdot \overline{R}/G$. Two choices in that expression are easy to get wrong and both matter. The time term must be the *busy* component, not the wall clock—using the wall clock charges idle time to the steps. And $\overline{R}$ must be the mean running count *during busy time*, not the mean over the whole run: the whole-run mean includes the idle samples, where the count is zero, so it equals $\overline{R}_{\text{busy}}\times$ (busy fraction).

Using it makes $\tau_1$ too low by exactly the busy fraction—36.3% for `cap1`, whose idle share is 36.3%. We report both columns in the data, and use the busy-conditional mean throughout; the whole-run mean is what an earlier revision of this paper used, and it made the two methods agree far better than they do.

*b) Method 2 (independent).:* For each step record we compute the per-request decode interval, $(\text{total_duration} - \text{first_token})/(\text{output_len} - 1)$, and take the median over steps with $\text{output_len} \geq 32$.

The two do not agree exactly, and the disagreement is informative rather than embarrassing. $\tau_1$ exceeds $\tau_2$ on every run, by 7.7–39.7% (`decomp.csv`, column `excess_pct`), which in absolute terms is 2.6–8.2 ms of a 27–37 ms step. The gap is *consistent with* engine steps that are prefill and therefore emit no output token—we can bound it, not identify it, because subtracting a median from a mean does not isolate a component. All we can form from these two is their ratio: the excess $\tau_1/\tau_2 - 1$ runs 17.7–39.7% across the five device-only configurations. Two things follow, and we separate them. First, it is a ratio of *times*, not of steps, and we have no measurement of the fraction of step *counts* that are prefill—chunked prefill makes those two different numbers. Second, even as a time ratio it is the gap between two measurements, not a decomposition of one: it is consistent with prefill and non-decode work filling it, but we cannot rule out that some of the difference is the two estimators disagreeing for other reasons. So the two are not two estimates of one number; they are measurements of two different things, and we do not size a component with their difference. Fig. 4 shows both, run by run. Method 2 remains the one that does not depend on the quantity it is used to explain, and it is the one we quote when the question is the cost of a decode step. The engine's own latency counters (`itl_p50`, `e2e_p50`, `ttft_p50`) are exported as zero by SGLang v0.5.10 in our configuration, so no directly measured step time exists in our data; instrumenting this is future work.

### C. The per-step cost is not constant in concurrency

Equation 3 would be uninteresting if $\tau$ were a constant. It is not. Across our 13 runs $\tau_1$ ranges from 27.0 to 36.7 ms and $\tau_2$ from 20.6 to 33.1 ms, and both rise with the running count: fitting Eq. 4 to the five device-only configurations gives $\varepsilon = 0.38$ for $\tau_1$ and 0.61 for $\tau_2$ ($R^2 = 0.89$ for the latter, Fig. 5).

The relation is a trend, not a law. Two host-tier configurations with similar running counts behave differently: `cap4_hc` runs 1.73 requests at 29.8 ms while `F:hz0` runs 1.66 at 35.7 ms. Running count sets a floor and a trend, not a value—and on the host-tier family, as we show below, it does not set the value at all.

This is the crux of the paper, and we state it at the strength the arithmetic supports. Hold $\tau_1$ at its value for `cap1`, 28.8 ms, and busy time falls as $1/\overline{R}$: raising the running count from 1.00 to 1.84 should have cut busy time by a factor of 1.84, from 340 minutes to 184. The measured curve flattens and lands at 234. The rise in $\tau$ takes back 50 of the 156 minutes

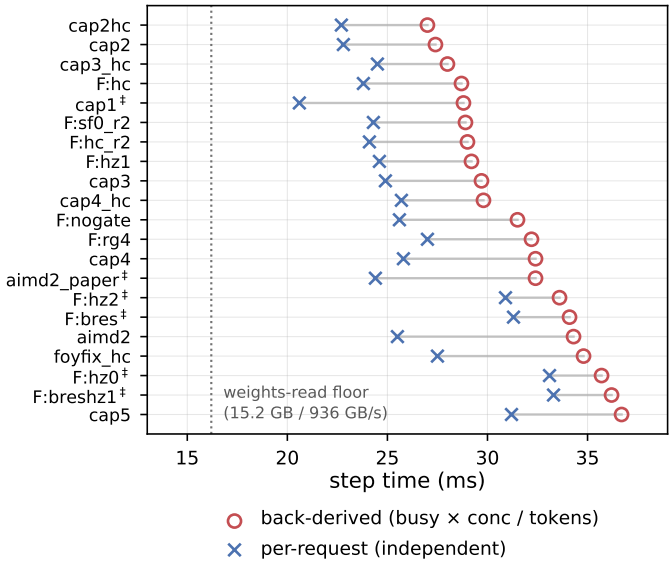



Fig. 4. Per-step cost by two independent measurements, one run per row: circles are model-consistent (all engine steps), crosses are the per-request decode interval. They do not agree—$\tau_1$ exceeds $\tau_2$ by 8–40%— and the gap is consistent with the prefill share, not measurement error; §III bounds that share rather than identifying it. The dotted line is the cost of reading the weights once; every run sits above it. The six rows marked ‡ were measured under host co-tenancy, which is why they cluster at the high end—their step times are the inflated ones (§IX-0e).

that the extra concurrency would otherwise have bought—*a third* of the gain, not all of it and, as an earlier revision of this paper claimed, not more than half. The other two thirds are real: raising concurrency does make the engine faster. What it does not do is make it faster in proportion, and the shortfall compounds with every controller that tries to exploit the difference.

*a) What the engine is doing when it is not decoding.:* SGLang exposes two counters that answer this directly: tokens evicted from the radix cache, and prompt tokens that had to be prefilled rather than reused. Across the five static caps *without* HiCache the two track each other almost exactly. `cap1` evicts 9.2 M tokens and prefills 8.6 M uncached; `cap5` evicts 21.3 M and prefills 20.7 M; the gap is 0.6–0.7 M on every one of the five. An evicted prefix is recomputed, close to one token for one, and that recomputation is chunked into decode steps. Those same five runs give recomputation rising with $\tau$ across four of the five points: 9.2 M uncached tokens at 27.4 ms, 11.5 at 29.7, 15.4 at 32.4, 20.7 at 36.7. The exception is the first point, `cap1`, which recomputes least (8.6 M) and still costs 28.8 ms—more than `cap2` does for more recomputation. We report the ordering we have rather than rounding it to a monotone one: it is a mechanism consistent with the $\tau$ trend on four of the five device-only configurations, and one point it does not order. It is five points on five different policies, not an intervention: we did not hold concurrency fixed and vary eviction, so we call it consistent rather than demonstrated.

*b) Where the mechanism stops.:* With HiCache enabled the correspondence breaks—and it breaks in a way that con-

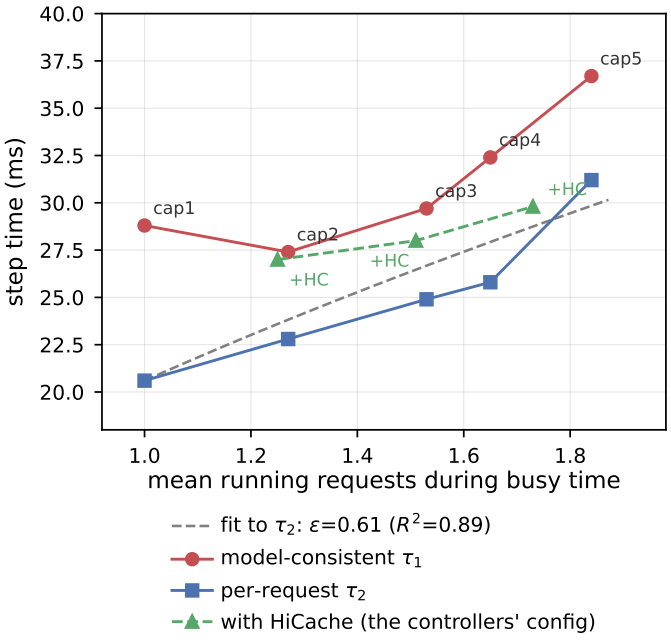



Fig. 5. Per-step cost against the running count during busy time. Solid circles are $\tau_1$ and squares $\tau_2$ for the five device-only configurations; $\tau_1$ is model-consistent (it averages over all engine steps, including prefill that emits no output token), $\tau_2$ is the per-request decode interval, and $\tau_1$ exceeds $\tau_2$ by 8–40%. The green triangles are $\tau_1$ for the three HiCache configurations—the ones the controllers run—and they are drawn against the same axes to make §IV's point visible: over $\overline{R}_{\text{busy}}$ 1.25 to 1.73, which covers the whole controller operating range, the host-tier curve rises 27.0 to 29.8 ms while the device-only curve over a comparable span rises 27.4 to 36.7. We plot $\tau_1$ only for the HiCache points to keep the figure readable. The fitted exponent is the elasticity $\varepsilon$ of Eq. 4; it is fit to the device-only points alone.

firms the mechanism rather than undermining it. The device cache misses as much as before: `cap4_hc` misses 22.4 M tokens on the device, more than the device-only `cap4` misses. But only 9.1 M of that is recomputed; the other 13.3 M is served from the host tier. Across the ten host-tier runs the host serves 12–19 M tokens each, and across the five device-only runs it serves *exactly zero*—the device miss equals the recomputed count to within rounding on all five. Recomputation is still what the device cannot absorb; the host tier is a second absorber in front of it.

### D. The tier, not the policy, decides whether reuse survives

The device-miss column hides the quantity that matters. Define *reuse* as the prompt tokens served from anywhere in the memory hierarchy—device radix cache plus host tier—so that reuse + recomputed = prompt, with the prompt fixed at 28.5 M by the trace.

The bound on reuse has to be defined independently of what we observed, or the argument is circular. We take it from the trace: summing the planned reusable prefix (`prefix_len`) over all 999 rounds of `replay_pois200_l04.csv` gives 21,540,371 tokens against a prompt total of 28,526,871, so no policy can reuse more than 75.5% of the prompt. That is a property of the workload. It is also a bound rather than a description: *no configuration we ran reaches it*, and the best realizes 92.6%, so the gap between "planned" and "achieved" exists even in the easiest case. What varies across configurations is how much of the bound is realized, and that is where the tier enters.

| | | | | | | |
|---|---|---|---|---|---|---|
| device-only | $\overline{R}_{\text{busy}}$ | 1.00 | 1.27 | 1.53 | 1.65 | 1.84 |
| | realized (% of bound) | 92.6 | 89.6 | 78.9 | 60.6 | 36.3 |
| | recomputed (M) | 8.6 | 9.2 | 11.5 | 15.4 | 20.7 |
| host tier | $\overline{R}_{\text{busy}}$ | 1.25 | 1.51 | 1.73 | | |
| | realized (% of bound) | 92.3 | 91.1 | 90.2 | | |
| | recomputed (M) | 8.6 | 8.9 | 9.1 | | |

Device-only, realization falls from 92.6% to 36.3% and the shortfall is recomputation: 8.6 M to 20.7 M tokens. With a host tier it holds at 90.2–92.3% on the static points that span the same range, and at 83.9–92.3% across the ten host-tier runs including the controller arms—the lowest of those being the arm that disables its own admission gate and therefore runs the pool hardest. and it holds while the *device* misses more than the device-only family's at every point (22.3–27.9 M against 8.6–20.7 M), because the host serves 13–19 M in front of the recomputation instead of letting the device keep it. A reader watching only the device hit rate would conclude the host tier made things worse. (The 92.6% ceiling-on-prompt and the 75.5% bound are not in competition: the first is what we measured, the second is what the trace permits; `reuse.csv` carries both columns and the raw counts.)

This is the same collapse §VIII reports under a different normalization. That section quotes the round-1-and-later prefix hit rate—steps restricted to rounds $\geq 1$, unweighted mean over steps—which falls from 80.8% to 32.8% across the device-only caps and holds at 80.7% to 78.7% with a host tier. The two conventions differ in two ways, not one: which steps are in the set (round $\geq 1$ against all rounds) and how steps are combined (unweighted mean against token-weighted sum). We use the token-weighted all-steps convention here because reuse and recomputation must add to the prompt, and we name the other whenever we quote it. Both are *realized* quantities that vary with the policy; neither is the trace's planned rate, which is fixed at 75.5% and appears in this section only as the bound.

Device occupancy does not explain the split. `cap3_fixed` and `cap3_hc` run at nearly the same mean device occupancy (55.8% and 55.5%) and nearly the same running count (1.53 and 1.51), and they realize 78.9% and 91.1% of the bound respectively. What differs is not how full the device pool is but what happens when it overflows: with no host tier an evicted prefix is gone and must be recomputed; with one it is demoted and read back. The pool is the binding resource, the tier sets the pool, and the tier therefore sets whether added concurrency is paid for in recomputation. The caveat is that this pair is two runs on two different days and two different GPUs, not a paired experiment; §IX-0e states what we did to rule out co-tenancy. A capacity-controlled version—same card, same trace, only the pool size changed—is queued behind the re-runs and we report it in §V when it lands; we do not use it

here.

*a) The second trace agrees on direction and not on level.:* We ran the same accounting on a second workload, reported in full in §VI—117 real Claude agent sessions, 390 rounds, the same pool and the same engine—and it reproduces the direction: its realization of *its own* bound falls from 74.8% to 67.5% as the running count rises from 1.08 to 1.29, a steeper per-unit drop than the device-only family's. What it does not reproduce is the level, and this is worth stating because it bounds what we can claim. That trace runs at a *lower* median device occupancy (15.8% against 22.4%) and realizes *less*, so realization is not a function of occupancy alone—and a statement of the form "the pool collapses above $x\%$ occupancy" is wrong on our own data. Occupancy is instantaneous while reuse depends on history; a session's prefix must survive from one round to the next, and how much pressure that puts on the pool is not visible in what the pool holds at an instant. We do not have the measurement that would separate those, which is why the controlled experiment we report in §V varies the pool alone, on one card and one trace, rather than leaning on the cross-trace comparison. The claim this section supports is the narrow one: within a configuration, the host tier preserves realization at a matched running count and a matched occupancy, and without it realization falls.

That gives the criterion of §IV a mechanism rather than a correlation, and it suggests a prediction we are testing: if the tier acts by multiplying usable capacity (the host pool is twice the device pool, so three times in total), then shrinking the device pool alone should move the collapse point even with no host tier involved. We do not treat the arithmetic as a result—it assumes KV demand grows linearly in the running count, which heterogeneous sessions need not satisfy—and the queued experiment measures the collapse point directly rather than predicting it, and §V reports it once it has run.

*b) What is left unexplained.:* Not everything here is accounted for. The host-tier runs sit at a narrow band of recomputed tokens (8.6–10.4 M) *and* of realization (83.9–92.3%), yet $\tau_1$ spans 27.0–36.2 ms across them. Part of that is now explained and part is not. Four of those runs (`F:hz2` 33.6, `F:bres` 34.1, `F:hz0` 35.7, `F:breshz1` 36.2 ms) were measured while three other engine instances shared the host, which inflates per-step cost by 25–40% and leaves the volume columns untouched (§IX-0e); they are being re-measured and we do not use their timing. Excluding them the remaining six span 27.0–31.5 ms at flat recomputation. That residual is small but it is not zero, and we do not attribute it. We note that the negative result does not rest on it: the cancellation is measured directly, not inferred from the mechanism.

One caution about the strength of the account above, since the arithmetic invites overreading. reuse + recomputed = prompt is a definition, not a finding, so "the shortfall reappears as recomputation" follows from how we partitioned the prompt and cannot by itself establish the mechanism. What is independent of the partition is the pair of engine counters: evicted tokens and uncached prefill tokens are separate SGLang counters, and on the device-only runs they track within 0.6–0.7 M on all five—which is the observation that licenses reading eviction as the cause of recomputation. With a host tier the two come apart by design, and the 13–19 M gap between them is the host's contribution.

*c) A refutation we had to withdraw.:* An earlier revision withdrew this mechanism outright, on two checks that were both wrong. The first was that total prefilled tokens equalled the trace's planned prefill to a ratio of 1.00, so no token could have been computed twice. `prompt_tokens_total` counts the full submitted prompt *including* its cached portion, so it equals the sum of prompt lengths by construction and cannot detect recomputation; the quantity that can is the uncached count used above, which varies by 2.4× across these runs. The second was that SGLang reported no retractions. Radix-cache eviction is not a retraction—retractions count running requests preempted mid-flight, a different event with its own counter. We record the episode because it is the same failure mode as the segment argument of §II: a check that looks like a measurement but is an identity.

### *E. Why idle time exists, and why recovering it does not pay*

The length of an individual idle stretch is a property of the workload, not of the controller. A session in a tool wait holds its permit and its KV but issues no request; with only about two live sessions, the probability that all of them are simultaneously waiting is substantial. The direct evidence is the duration of the stretches themselves: they are of the same order as the tool waits, and an order of magnitude longer than the 2 s control cycle (Fig. 3). No controller can shorten a gap that the tool set.

How much idle a *run* accumulates is a different matter, and there the controller is not off the hook. Idle accumulates when every live session happens to be parked at once, and how many sessions are live is precisely what an admission policy sets. Two configurations with the same mean running count make this visible: `horizon`=1 (1.52) and `cap3`+HiCache (1.51) idle 36.7 and 24.1 minutes respectively—a 52% difference at matched concurrency—and 12.6 of the 19.4-minute wall-clock gap between them is that idle difference rather than step cost. So the gap durations are the workload's, and the number of gaps is partly the policy's.

We note one claim we had to drop here. An earlier draft reported that idle fraction falls monotonically with mean concurrency. With the full data it does not: `F:hz0` at 1.41 running requests idles 15.2% while `cap4` at 1.48 idles 10.4%. The weaker statement that survives is the one we use—the stretches are structural and long.

The obvious remedy—keep more sessions live so that someone is always ready—is what `bres` does. It works in the narrow sense: `bres` recovers about 17 of the minutes that its own baseline spends idle. It is still slower overall, because the sessions it keeps live raise the step cost by more than the recovered idle is worth—and, as above, that step-cost rise is measured under co-tenancy and is therefore an upper bound. Idle is not unrecoverable; it is recoverable at a price that is not worth paying here. The structural remedy would be to make

the engine do work during a wait, but there is no work to do: the session's next round does not exist until its tool returns.

## IV. A Criterion for When Admission Control Can Pay

The decomposition turns the question into one number. Throughput during busy time is $\overline{R}/\tau$, so raising concurrency by a factor $f$ buys a factor $f^{1-\epsilon}$, where

$$\epsilon \;=\; \frac{\mathrm{d}\ln\tau}{\mathrm{d}\ln\overline{R}} \tag{4}$$

is the elasticity of per-step cost with respect to the running count. At $\epsilon = 0$ the step cost is flat, concurrency converts one for one into throughput, and an admission controller has room to work. At $\epsilon = 1$ the two cancel exactly and it has none. Everything in between is a quantitative statement about how much room.

On the five device-only configurations, where §III establishes that the step cost is driven by recomputation, $\overline{R}_{\text{busy}}$ runs from 1.00 to 1.84. A least-squares fit of Eq. 4 over those five points gives $\epsilon = 0.61$ on the per-request measurement ($R^2 = 0.89$), or $0.38$ on the model-consistent one (Fig. 5).

That fit is a straight line through a curve, and the curve is what matters. The segment elasticities are $\epsilon = -0.21$ from `cap1` to `cap2`, $+0.43$ from `cap2` to `cap3`, $+1.15$ from `cap3` to `cap4`, and $+1.14$ from `cap4` to `cap5`. Below $\overline{R}_{\text{busy}} \approx 1.5$ added concurrency still buys something; above it, $\epsilon$ passes one, which means the engine gets *slower* in wall clock as more requests run. The single number $0.38$ averages those regimes and describes none of them.

The operating region is the one that matters, and it is not the region the controllers run in. All nine controller arms run with HiCache enabled, which the five device-only caps above do not. Measuring the same elasticity on the three HiCache configurations we have gives $\epsilon = +0.19$ from `cap2`+HC to `cap3`+HC and $+0.46$ from `cap3`+HC to `cap4`+HC, over $\overline{R}_{\text{busy}}$ 1.25 to 1.73. That range covers the whole controller operating region (1.35 to 1.69), and over it $\epsilon$ never approaches one. At matched running counts the HiCache caps pay 1.7 ms less per step at $\overline{R}_{\text{busy}} \approx 1.5$.

This contradicts what an earlier draft of this paper claimed, and it changes what the criterion is for. The cliff in the device-only curve is a recomputation effect: §III measures that those runs recompute what they evict, while under HiCache the host tier absorbs it. The cliff does not survive on the configuration our controllers actually use. So the criterion does *not* explain why they lose—on their own configuration they were not in a region where added concurrency was self-defeating, and the engine had room they did not take. What the measurement does is rule one explanation out. The failure is not that this workload denies admission control its lever; it is that our controllers do not pull the lever they have, which is what §V-B and the attribution of §V-C measure directly.

We are explicit about what this does and does not establish. The HiCache curve rests on three configurations and two segments, one run each, with no error bars, and it stops at 1.73—we cannot see whether a cliff appears beyond the controllers' region. A $\pm 10\%$ perturbation of $\tau$ puts the segment elasticities' bounds across one even on the device-only curve, so the cliff is a description of the fitted points rather than a threshold we have located precisely. And it is measured on different policies rather than by an intervention, so it is a cross-policy association. On our data it is also computed from the same wall clock the model then explains—$\tau_1$ is derived from $W_{\text{busy}}$, so using it to predict $W_{\text{busy}}$ restates rather than tests—and its points are different policies, so the slope is a cross-policy association, not a transportable causal derivative. $\tau_2$ is genuinely independent of the wall clock but measures only decode, so it cannot be dropped into a throughput equation that includes prefill.

We tested whether the curve transfers to a second workload and it does not obviously do so. On a real agent trace (117 sessions, median inter-round waits of 20.2 s against 0.34 s on the synthetic trace, `toolwait.csv`, three static caps and one controller arm), the model-consistent elasticity is negative across the measured range—$-2.40$ then $-0.28$, at $\overline{R}_{\text{busy}}$ 1.08 to 1.29—and the controller arm there runs at 1.22, below every one of the nine arms above, and still does not beat the static front. The per-request measure has the opposite sign on that trace ($+0.41$, $+0.48$). We report this as a boundary: the criterion is a description of one workload on one configuration, and where we could check it elsewhere it neither transferred nor explained the failure. It is not, as an earlier draft claimed, something that decides the question in minutes before building anything.

## V. Results: Four Routes, One Outcome

### A. The frontier

Fig. 1 places every completed run on the (wall clock, SLO) plane and draws the upper-left envelope of the static points. Table I gives the numbers and each controller's signed distance from that envelope.

The result is unambiguous. Of nine controller variants, *none* is above the envelope—not one, on any run, and the pair we repeated most tightly says so at a resolution the others do not. The closest arm of all is the one with single-flight admission disabled, at $0.55$ percentage points *below* the line—inside our $\pm 1.0$ pp tie band, so we record it as *on* the line rather than below it, but on the below side of it.

The base configuration is the one whose repeats we can trust, and they are tight: four runs spanning 0.5 pp of SLO (89.5% mean, 279.8 to 293.2 minutes), against the 3.3 pp that separates it from the single-flight-off arm. So that difference is several times the noise of its own configuration rather than inside it, and it is a trade rather than a win: 28 minutes of wall clock bought with 3.5 pp of SLO, moving the arm onto the tie band from below it. Not one configuration in the table gets above the line, and the tightest pair we have is what makes us willing to say the near misses are misses. A fifth pair, two runs of `cap3`+HiCache, *is* used, and an earlier draft of this paper wrongly set it aside saying one of them exceeded its own cap. That was a different run: the over-cap one is the device-only

cap3 repeat (peak in-flight five against a cap of three; at our 300-point in-flight sample grid, 126 samples sit over the cap, a count that scales with the grid rather than with the run), while both HiCache runs peak at three with none. The repeat is also both faster and higher on SLO (230.3 minutes, 81.9% against 236.4 and 81.1%), so by the same rule that displaced cap4+HiCache's first run it displaces this one—and moving that anchor is what puts our best arm below the line rather than on it.[1]

The anchor repeats moved the frontier twice and we report both moves. The leftmost anchor's second run came out both faster (209.1 against 213.0 minutes) and higher on SLO (68.5 against 63.4), displacing the first; no controller verdict moves, because every controller finishes to the right of the next anchor. The middle anchor's repeat moved it the other way for us: it is also faster and higher, and taking it moves our best arm from *on* the line to *on the line, below it*. We report every repeat we ran rather than keeping the better of each pair, because a frontier whose anchors are single measurements is the weakness this whole subsection is about—and here it is the weakness biting.

The other end of the front deserves its own sentence, because the rest of this paper is negative and a reader could close it thinking the answer is to do nothing. The run with *no* admission control—every session admitted the moment it arrives—finishes in 288.6 minutes at 3.0% SLO, 87.6 pp below the envelope and the worst SLO we report. Its median time-to-first-token is 1,774 seconds, its prefix cache hit rate collapses from the planned 75.5% the runner prints in its end-of-run summary (the all-steps convention, a different one from the round-1+ numbers in §VIII—for the same trace gen_table.py reports 73.8% on that convention) to 1.6%, and its segment SLO is 30% in the first tenth of the run and exactly zero in each of the nine after that. Uncontrolled admission does not degrade gracefully; it destroys the cache and then serves almost nobody. The question this paper addresses is therefore not whether to admit selectively, but which selective rule to use.

Two features of the front are worth noting before the negative results. The envelope is steep between 209 and 230 minutes (0.63 pp per minute) and shallow thereafter (0.15 pp per minute), so the cost of buying SLO by retreating from the fast end is high. And $W$ is not monotone in concurrency: five sessions (260.9 min) is *slower* than four (249.9 min)—the same configuration family, no host cache on either side—because the per-step cost rises faster than the request count does. We return to this non-monotonicity in §III, where it turns out to be the first appearance of the mechanism that drives the whole frontier.

[1]The repeats are uneven, and we do not want the tightest pair to stand for all of them. Five configurations have been run more than once; their SLO spreads are 0.5, 0.8, 1.0, 3.9 and 5.1 pp (replicates.csv). The two widest belong to the two arms that decide something—horizon=1 and the leftmost anchor—so where a small margin matters our repeats are weakest, and where they are strongest (the base configuration and the cap3+HiCache anchor) the margin is not small. We report both ends rather than the average.

TABLE I

ALL COMPLETED RUNS. *Conc.* IS THE TIME-WEIGHTED MEAN NUMBER OF *live* SESSIONS, WHICH IS NOT THE RUNNING-REQUEST COUNT $\overline{R}$ USED IN §III—A LIVE SESSION IS PARKED IN A TOOL WAIT MOST OF THE TIME, SO CAP5 SHOWS 4.89 LIVE AGAINST 1.84 RUNNING. $\Delta$ IS THE SIGNED DISTANCE FROM THE STATIC ENVELOPE AT THAT WALL CLOCK; ABOVE $+1.0$ PP COUNTS AS A WIN, BELOW $-1.0$ PP AS A LOSS, AND THE BAND BETWEEN AS A TIE (§II). FOUR RUNS STOPPED SHORT OF 999 (HZ0 995, HZ2 997, RG4 998, AND THE NO-CONTROL BASELINE 997); THEY ARE LISTED WITH THEIR MEASURED VALUES. WE FLAG THIS BECAUSE OUR VERDICT SCRIPT EXCLUDES RUNS BELOW 999 STEPS, SO THE $\Delta$ COLUMN FOR THESE FOUR COMES FROM THE TABLE GENERATOR RATHER THAN FROM THAT SCRIPT. ‡ MARKS A RUN MEASURED WHILE THREE OR MORE OTHER ENGINE INSTANCES SHARED THE HOST FOR AT LEAST AN HOUR. THAT IS A RISK FLAG, NOT A MEASUREMENT OF DAMAGE, AND THE TWO COME APART IN OUR DATA: FOUR OF THE SIX FLAGGED ROWS ARE CONTROLLER VARIANTS WHOSE PER-REQUEST DECODE INTERVAL IS 30.9–33.3 MS AGAINST 23.8–25.7 FOR THE SAME FAMILY MEASURED ALONE, SO THEIR WALL CLOCK, SLO AND $\Delta$ ARE NOT COMPARABLE WITH THE REST AND THEY ARE BEING RE-MEASURED. THE OTHER TWO, CAP1 AND THE PUBLISHED-PARAMETER AIMD ROW, ARE FLAGGED BY THE WINDOW BUT SHOW NO AGGREGATE INFLATION AT ALL (20.6 AND 24.4 MS), SO WE KEEP THEIR NUMBERS AND SAY WHY. §IX-0E GIVES THE AUDIT, THE THRESHOLD AND THE EVIDENCE.

| Policy | Wall (min) | Conc. | SLO (%) | $\Delta$ (pp) |
|---|---|---|---|---|
| *Static caps* | | | | |
| no control | 288.6 | 1.69 | 3.0 | −87.63 |
| cap1‡ | 527.4 | 1.00 | 94.8 | *anchor* |
| cap2 | 304.2 | 1.99 | 89.5 | −2.23 |
| cap3 (fixed) | 258.7 | 2.89 | 74.4 | −11.77 |
| cap4 | 249.9 | 3.94 | 55.7 | −29.17 |
| cap5 | 260.9 | 4.89 | 27.1 | −59.34 |
| *Static caps + HiCache* | | | | |
| cap2+HC | 295.0 | 1.99 | 91.6 | *anchor* |
| cap3+HC | 236.4 | 2.93 | 81.1 | −1.72 |
| cap3+HC (repeat) | 230.3 | 2.97 | 81.9 | *anchor* |
| cap4+HC | 213.0 | 3.93 | 63.4 | −7.55 |
| cap4+HC (repeat) | 209.1 | 3.87 | 68.5 | *anchor* |
| *Foyer controller variants* | | | | |
| single-flight off | 263.3 | 2.43 | 86.3 | −0.55 |
| horizon=1 | 255.8 | 2.58 | 84.8 | −0.93 |
| horizon=3 (base) | 291.3 | 2.06 | 89.8 | −1.24 |
| horizon=3 (2nd run) | 293.2 | 2.07 | 89.6 | −1.73 |
| horizon=2‡ | 306.1 | 2.22 | 86.9 | −4.84 |
| horizon=1 (repeat) | 260.8 | 2.69 | 80.9 | −5.58 |
| resident base‡ | 295.5 | 2.40 | 83.8 | −7.81 |
| horizon=0‡ | 290.0 | 2.69 | 77.2 | −13.64 |
| horizon=1, no hw‡ | 310.9 | 2.77 | 72.3 | −19.54 |
| no gate, no resv. | 272.3 | 3.25 | 66.6 | −21.62 |
| running gate=4 | 282.3 | 3.03 | 64.3 | −25.38 |
| all throttles off | 415.0 | 2.58 | 46.1 | −47.10 |
| *Independent adaptive controller (§V-G)* | | | | |
| AIMD, paper params‡ | 364.3 | 1.91 | 81.4 | −11.16 |
| AIMD, retuned | 264.6 | 3.87 | 52.8 | −34.28 |

### B. The controllers leave memory unused

Before the routes, one measurement that reframes all of them. Our controller keeps every session's permit for its lifetime, so its pool occupancy is a policy output, not a constraint. It runs at a median of 49.2% of the pool on the base configuration, and the median number of sessions queueing for admission is 66. The configuration that finishes fastest—the

repeat of that same static cap of four, at 209.1 minutes—runs at 71.9%, and its first run at 70.2%. Occupancy and running requests track each other reliably across the arms we have (the two columns in `occupancy.csv` rank the twenty arms almost identically), but running count does *not* order wall clock: `horizon`=2 runs 1.25 requests against the base arm's 1.14 and finishes 15 minutes slower. That gap is what the next subsection is about.

The controller believes it is nearly full. Its own decision log records the projection it admits against—the largest of measured occupancy and the summed contexts of live sessions, plus a reserve for their growth—at a median of 89,073 tokens against a budget of 96,360, i.e. 7% free. Its measured occupancy at the same instants is 49,875 tokens: 48.2% of that budget, or 50.8% of the pool, is idle. The projection is $1.79\times$ the truth on this configuration. Across the controller runs we have projection logs for, the ratio runs from 1.24 to 2.00 with a median of 1.75 on this trace, and from 1.24 to 4.11 with a median of 1.75 once the second trace's controller arm is included (`projection.csv`; the second trace is the 4.11). So 1.79 is this arm's value, not a constant of the design, and we name the set because the range depends on it—over every run in that file, including the flash-crowd family and the 25-session tier, it is 1.01 to 4.11 with a median of 1.35. Two terms produce the gap: charging every live session its full context, when a parked session's KV is evictable and therefore absent from the engine's count, and the growth reserve, which alone is 26,762 tokens, 28% of the budget.

Two things follow, and we are careful about the second.

The first is that the controllers under-use the pool, and that this is a property of their estimates rather than of the hardware: the base configuration holds nearly half the pool idle (48.2% of its budget, 50.8% of the pool) with 66 sessions queued, because it believes it has 7% free when it has 48%. Any account of why it is slow that does not mention this is incomplete.

We then asked what actually stops it, by instrumenting the admission decision. The path is a conjunction: a session is authorized only if the SLO valve is open, the round gate is not full, and its projected footprint fits what remains of the budget. Over the base run's 8,196 blocked cycles—cycles with a session waiting and none admitted—the valve accounts for 6.8%, the round gate for none, and a projection already above budget for 38.4%.

The remaining 54.8% are cycles where the controller selected no candidate at all, and here we are explicit about the limit of what we can resolve. The controller is single-flight by default: it authorizes at most one session per two-second cycle and waits for that session to begin a round before authorizing the next. Whether that gate held or whether a candidate was tried and did not fit is *not distinguishable from our logs*—the controller does not record it—so we do not attribute the bucket to either. We bounded it by running that arm: the same configuration with single-flight disabled. Two things came out of it, and the second is the one that matters.

The bound first. With the gate off, the residual bucket—which can then only be *candidates that did not fit*, since the gate cannot be holding—falls from 54.8% of blocked cycles to 39.7%, against 44.0% that are over budget outright. So the single-flight path accounts for roughly fifteen points, and the headroom estimate remains the largest single cause. (Both arms' numbers are rows of `gate_attrib.csv`, which now includes the `sf=0` arm; an earlier revision quoted 39.9 and 43.5 for the second pair, values that appeared in no data file.)

The second thing is that disabling it does not make the controller win. `sf=0` finishes in 263.3 minutes at 86.3% SLO against the base run's 291.3 and 89.8%: it buys 28 minutes for 3.5 pp, and its signed distance from the front moves from $-1.24$ to $-0.55$—onto the tie band from below it, not above it. Its admission rate is unchanged (2.5% of cycles admit, against 2.3%). So the gate was a real cost and removing it moves the arm along the front rather than past it, which is the same trade every other route in this section makes.

Across the eight Foyer variants that left controller logs—the set is named in `gate_attrib.csv`, and it is eight rather than nine because `foyfix` and the two AIMD arms run a different controller and log no gate state—the projection is over budget in a median 32.2% of blocked cycles (19.5% to 50.8%) and the valve in a median 32.7% (6.8% to 67.9%). Both medians move with the set, which is why we name it: over every run in that file they are 27.0% and 5.4%. One convention note, because it moves these numbers: we take the median as the upper of the two middle values when the count is even (`v[len(v)//2]`, which is what `gen_gate_attrib.py` computes and what the file's `eight_median` line records), not the mean of the two; on eight arms that difference is 0.5 pp on the projection and 6.7 pp on the valve. The round gate never binds except in the two runs that enable it—`rg2` at 29.3% and `rg3` at 18.0% of blocked cycles, against 0.6% or less everywhere else—which is what one expects of a gate that is off by default. We also record what this accounting omits, because it changes the picture: the controller has a starvation guard that bypasses the budget entirely once the oldest waiter has waited 240 s, cooling down for 300 s. On the base run it accounts for 24.8% of all admissions (51 of 206), and while the projection is over budget it is the *only* way in—of 3,700 such cycles, nine admitted, and all nine were forced. It is therefore the guard, not the budget test, that sets the floor on how fast this controller can admit.

The second is that under-use does *not* order the outcomes, so it is not the whole explanation either. `horizon`=1 and `cap3`+HiCache sit at the same median occupancy (57.8% of the pool) and the same mean running count (1.52 against 1.51 on the busy-time measure of §III), and the cap is 19 minutes faster. `hz0` runs at 64.1%, higher than that cap, and is 54 minutes slower. Occupancy buys running requests reliably—the two columns in `occupancy.csv` track each other across all twenty arms—but converting running requests into wall clock is not simply the per-step cost of §III. Decomposing that 19 minutes into its idle and busy components, 12.6 of them are idle time—65%—and only 6.8 are busy; the two runs' step costs differ by 1.2 ms (29.2 against 28.0), about 4%. A

parked session is one the controller has authorized but not yet released, so the same policy that sets the running count also sets a large part of the idle time. Both effects are real; on this workload the unused-memory one is larger for the arms that finish worst, and the step-cost one is what separates the arms that finish close together.

### C. What the two mechanisms explain, arm by arm

We have offered two mechanisms, and the honest thing is to say what each one covers. For every controller arm we took the static cap closest to it in wall clock and compared their occupancy and their step cost. An arm is attributed to the unused-memory mechanism if it runs fewer than 92% of its comparison's requests, and to the step-cost mechanism if it runs a comparable number—within 8%—at a measurably higher cost (more than 8% higher $\tau_1$). Everything else is a residual. The rule reads the running count and $\tau_1$ only; it does not consult occupancy, so the "unused memory" label is licensed by the running count alone—though the two arms it lands on also sit at the low end of the occupancy column, which is why we let the name stand. The comparability clause is load-bearing and we state it because an earlier version of this table did not enforce it, which classified every arm that ran *more* requests than its comparison as step-cost purely because $\tau$ rises with the running count.

| Arm | Occ. | $\overline{R}$ | $\tau_1$ | Attribution |
|---|---|---|---|---|
| single-flight off | 54.5 | 1.455 | 28.9 | unused mem. |
| `horizon=1` | 57.8 | 1.524 | 29.2 | residual |
| Foyer base | 49.2 | 1.353 | 28.7 | residual |
| `horizon=2` | 54.7 | 1.439 | 33.6 | $\tau$ high, not matched$^{\ddagger}$ |
| `horizon=0` | 64.1 | 1.666 | 35.7 | $\tau$ high, not matched$^{\ddagger}$ |
| `bres` | 56.9 | 1.481 | 34.1 | $\tau$ high, not matched$^{\ddagger}$ |
| `breshz1` | 61.4 | 1.632 | 36.2 | $\tau$ high, not matched$^{\ddagger}$ |
| no gate, no resv. | 62.1 | 1.678 | 31.5 | unused mem. |
| running gate=4 | 62.3 | 1.689 | 32.2 | $\tau$ high, not matched |
| all throttles off | 39.2 | 1.675 | 34.8 | $\tau$ high, not matched |
| AIMD, retuned | 66.6 | 1.706 | 34.3 | residual |
| AIMD, paper params | 40.3 | 1.357 | 32.4 | step cost$^{\ddagger}$ |

$^{\ddagger}$ marks runs measured while three or more other engine instances shared the host for at least an hour (§IX-0e). In this table the flag bites harder than in Table I, because the classification is made on $\tau_1$, which is exactly the column co-tenancy inflates. Four of the six $\tau$-high rows carry it, and so does the one step-cost attribution the procedure makes—AIMD at its published parameters. That row is flagged by the window but not measurably damaged: its $\tau_2$, which does not touch the sampler, is 24.4 ms, inside the clean family's range. We report it as flagged rather than calling it clean, because the window is what the audit establishes and the absence of damage is an inference from a different column. The two $\tau$-high rows that are not flagged (`running gate`=4 is a watch, all-throttles-off is clean) are the ones whose elevation no window explains.

3 of the 12 rows are attributed: two to unused memory (`single-flight off` and no-gate-no-reservation), one to step cost, six have a $\tau$ no matched comparison can confirm, and three are residuals—their occupancy and their step cost both sit close to their comparison cap's, and their shortfall is real but we cannot say from these measurements which mechanism produced it. Two limits on what that buys, both of which we want read before the number is quoted. First, the comparison is a static cap matched on *wall clock*, and the arms that finish near 260 minutes are matched against `cap5`, whose SLO is 27.1%—so the classification describes an arm relative to a wall-clock peer, not relative to the static *front*, and it cannot say why the arm fails to beat the front. Second, the two unused-memory rows run fewer requests than their comparison, which is a description of what they did, not a demonstration that doing otherwise would have won; §V is the section that establishes the loss, and this table only sorts it. We publish the table rather than a summary because a summary would claim more than we have—and because "two mechanisms explain the negative result" would be the wrong summary. We also flag what changed here, since an earlier version of this table claimed four attributions from eleven rows and that was an artifact of two errors at once: the comparability clause above was not enforced in the code, and six arms had no $\tau_1$ because the run list behind this table was incomplete. The generator is now the only source for these rows (`attribution.csv`), it enforces the clause, and it carries the co-tenancy flag rather than having it written in by hand—which is how the earlier version came to mark two clean runs.

### D. Route 1: the accounting base

Our controller's projection (Eq. 1) charges a session for its full context even while it is parked in a tool wait, when its KV is evictable and therefore absent from the engine's reported occupancy. We argued this was a double charge and removed it, admitting on engine telemetry alone.

The charge is not double. Removing it did admit more—median engine occupancy rose from 49.2% to 56.9% of the pool—and the round-1+ prefix hit rate held at 80.2% against the 80.6% of the run that keeps the charge, so no eviction spiral occurred. But the run was 4.2 minutes *slower* and lost 6.0 pp of SLO. Per §III, the extra occupancy raised both $\overline{R}$ (1.35 to 1.48) and $\tau_2$ (23.8 ms to 31.3 ms), and the two cancelled.

That last sentence is the one place where this route's conclusion depends on a contaminated measurement, and we mark it. The arm that tests this route (`bres`) is one of the four flagged $^{\ddagger}$ in Table I: its 31.3 ms $\tau_2$ was measured under host co-tenancy, and the same arm measured clean would plausibly sit near 25 ms rather than 31. On that reading the two terms would not cancel and the route would look better than we report, not worse—so the direction of the correction is against our own conclusion, which is the safe direction to be wrong in, but it is still unresolved. The occupancy half of the argument (49.2% to 56.9%) is a volume measurement and is unaffected. We keep the route in the negative column pending the re-run and say so here rather than in a footnote.

We note that this claim had already been falsified once by source inspection: SGLang's `token_usage` subtracts

evictable blocks, so a parked session's prefix is genuinely not counted, and charging for it reserves capacity that really will be needed back. The end-to-end measurement agrees with the source reading.

### *E. Route 2: the growth reserve*

Sessions grow by roughly 2.5K tokens per round, so the controller reserves capacity for the next `horizon` rounds. We swept the horizon over $\{0, 1, 2, 3\}$ and the reserve is not a free parameter to be minimised: Fig. 6 shows the response is non-monotone. Removing the reserve entirely costs 12.6 pp of SLO, and a reserve of two rounds is 15 minutes *slower* than three. The optimum is at one round, and there it comes closest to the static front—0.93 pp below it, inside the tie band—and that single point is the one this paper spends the most words on, because it does not reproduce: the repeat of this configuration lands 5.6 pp below the line (§V). Read the sweep as a shape, not as a result at its minimum.

### *F. Route 3: the hard ceiling*

Lowering the budget fraction is the crudest available knob and behaves as the other routes do: it slides the operating point along the front. We report it only for completeness, since it adds no mechanism.

### *G. An independent adaptive controller*

Every controller above is ours, which invites the objection that the negative result is a statement about one design rather than about adaptive admission. We therefore implemented a second, mechanistically different controller: an AIMD window over an agent-level admission set, following the control law of the congestion-based system we compare against in §VIII. The window grows additively while cache occupancy sits below a low watermark, is cut multiplicatively when occupancy exceeds a high watermark *and* the prefix hit rate falls below a threshold, and holds otherwise. Cuts pause the youngest sessions at their next step boundary; growth resumes paused sessions first. Every input is a runtime gauge—occupancy and hit rate from the engine, nothing from the trace—and the window acts through the same named permits as Foyer.

It does not beat the static front either, at either parameter setting (Table I). At the values reported in that paper's evaluation it finishes in 364.3 minutes at 81.4% SLO, 11.2 pp below the envelope; the same arm with its retuned watermarks finishes in 264.6 minutes at 52.8%, 34.3 pp below (`table1.csv`). The retuned setting is also one of the most volatile runs we report: its segment-to-segment SD is 13.2 pp of SLO, with segment values swinging between 37 and 72. Three runs are worse (`cap4` at 15.8, the all-throttles-off arm at 14.4, and `breshz1` at 14.2), and no other controller arm is as bad—which is what a control loop oscillating around a capacity cliff looks like.

We report two things honestly about this comparison. First, this is our reimplementation of a published control law, not the authors' system, and a reimplementation of any controller can underperform its paper. We built it to the published law, gave it the same permit interface and the same 2 s cycle as Foyer, and did not tune it beyond the two documented settings; but the caveat stands. Second, both controllers stall. At its paper parameters this one pinned its window at one session, admitted nothing, and left the engine idle for over an hour. Our own controller does the same thing more briefly: in 4 of the 11 Foyer runs an audit found a single window of 150–175 s in which the engine was empty, the queue held 90 sessions, the SLO valve was open, and the controller issued no permit at all—about 0.9% of each run's wall clock. The remaining runs, including the best one, show no such window. Nothing is logged as an error and the run recovers on its own.

We traced one such window in a repeat of our best configuration and found a code path that explains it: single-flight admission considered only the head of the FIFO queue, and when the head was a session too large to fit the remaining budget it was never admitted, never left the queue, and blocked every smaller session behind it. That is the code every run in this paper executed. The fix in our repository—scan the queue, still admit at most one per cycle—is *not* in the artifact that produced these results, so the numbers here predate it and we do not claim them for it. We are also explicit that this is *a* path, not a proven cause: the audit is a log signature, and the same signature can be produced by other closed gates. What we can say is that the signature is present in a third of our Foyer runs and absent from all of our static caps, which have no admission decision to get stuck in.

That is the honest version of a finding we first reported too strongly. An intermediate draft of this paper called it a deadlock and said the configuration that produced our best result was the one that deadlocked on its repeat. Neither holds: the windows are 160 s and self-clearing, and the best configuration never showed one. The windows are also too small (0.9%) to move any verdict we report. We keep them because the mechanism is real and its size is the point: on a pool that holds four median sessions, a controller's failure mode is a silent stall, not graceful degradation, and it took a repeat run to see it.

### *H. Route 4: execution granularity*

The first three routes all assume that what a controller can set is *how many sessions are resident*. A static cap sets the same quantity, so it is unsurprising that they land in the same place. The fourth route attacks the assumption instead: because a large fraction of live sessions are parked at any instant, a cap on *resident* sessions leaves the engine idle whenever all residents happen to be parked. Capping the *running-request* count instead should keep the engine busy while leaving the resident count high.

The permit design supports this directly: a paused session finishes its current round and blocks before its next one, and resumes when the controller re-admits it. We implemented it as a gate on the engine's own `num_running_reqs` and swept the gate over $\{2, 4\}$. Only the gate-of-four arm completed (999 steps); the gate-of-two arm reached 770 and is not used, so

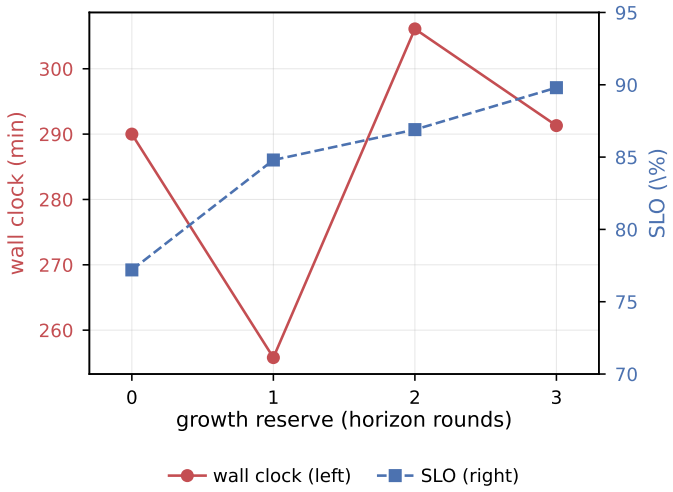



Fig. 6. The growth reserve is not a free parameter. Removing it costs 12.6 pp of SLO; two rounds is 15 minutes slower than three. The optimum at one round comes closest to the static front, 0.9 pp below it—and is the one point here that does not reproduce on a repeat (it lands 5.6 pp below). The other three settings were each run once, so the shape is better supported than any single point on it.

what we report below is a single comparison against the no-gate arm rather than a sweep.

It does not work, and the measurement says why. Against the no-gate arm, a gate of four moved mean running requests from 1.33 to 1.30 and mean live sessions from 3.25 to 3.03: it tightened both, by 2% and 7%, rather than pinning the one it aims at. That is what we observe, and we state it as the observation it is rather than deriving a bound we cannot support. A live session spends most of its time parked in a tool wait (the ratio of mean running requests to mean live sessions is 0.38 to 0.66 across our arms), so pausing live sessions removes capacity that was not running in the first place—but that ratio is an average over a run, not a bound on any instant, and the running count can in principle reach the live count. Both repairs fail: without a constraint on admission, a paused session is re-admitted on the next cycle and blocks for at most one 2 s cycle; with one, the policy is a session cap again. The obstacle is granularity—`paused` takes effect at round boundaries, while pinning an instantaneous count requires control the permit mechanism does not have.

The measured outcome matches: the gate moved running concurrency from 1.33 to 1.30 and live sessions from 3.25 to 3.03. It tightened rather than pinned, and a tightening is another cap.

## VI. The Same Measurements on a Second Workload

Everything so far is one trace family, and a negative result drawn from one family invites the objection that the family is the finding. We ran the same accounting, the same static sweep and the same controller on a second replay: 117 real Claude agent sessions, 390 rounds, the same model, the same engine and the same 101,432-token pool. One difference has to be stated up front, because it bears on the paper's central variable: *this trace was run entirely without a host tier*. Its static points, its envelope and its controller are all device-only. That makes the comparison internally fair—both sides of the front are the same configuration—but it means this workload tests the negative result and not the tier mechanism, and it leaves an alternative account of the controller's failure standing: it failed here on the configuration where §III says added concurrency is expensive. The two workloads differ in ways that matter for this paper, which is why we chose it. Its sessions are smaller (median prompt 18,809 tokens against 23,187) and its tool waits are far longer at the median: over round transitions—same session, adjacent rounds, each session's last round excluded because it has no successor—the gap is 20.2 s against 0.34 s at the median—about 60×—and 39.9 s against 15.8 s at the mean, which is 2.5× (`toolwait.csv` carries both traces on this definition). The 29.1 s that appears in `elasticity_x.csv` is the mean over all 390 rows including each session's last round, and is not comparable with the main trace's 15.8 s; we use the transition definition throughout §VI. So the engine is idle far more of the time, and the running counts are lower.

### A. What transfers, and what does not

*a) The reuse accounting transfers in direction, not in level.:* The second trace plans a higher reuse bound than the first—86.3% of its prompt tokens against 75.5%—and realizes less of it, at a lower device occupancy. Realization falls from 74.8% to 64.0% as the running count rises from 1.08 to 1.41, so the direction of §III survives. What does *not* survive is the shape. On the main trace the decline is a threshold: the per-unit slope is $-10.8$, $-41.0$, then $-157.4$ and $-126.7$ points per unit as the running count passes 1.53, which is the cliff §IV measures. Here the slope is roughly constant—$-31.3$, $-36.6$, $-28.4$—over a span that reaches 1.41, so at 1.41 this trace has lost more of its bound than the main trace had at 1.53 (36% against 21%) and has not yet begun to fall off a cliff. The cliff is a threshold that this trace has not reached at the running counts its own offered load produces; whether it has one at all, and where, is what the rest of the sweep is for. We do compare the two traces' overall slopes, which is the one slope comparison the data supports because both are taken over their full measured ranges (`slope.csv`): the device-only family runs from 92.5% to 36.3% over $\overline{R}$ 1.000–1.842, a whole-range slope of $-66.8$ points per unit, and this trace from 74.8% to 64.0% over 1.076–1.409, a slope of $-32.3$. So per unit of running count, the main trace loses its bound twice as fast over its range—but the comparison is between ranges of different length and different absolute level, and we do not lean on it. The claim we do lean on is the shape one below. The level does not, and we treat that as a bound on what we can claim: this trace realizes *less* at 15.8% median device occupancy than the main trace does at 22.4% (both medians of `sglang:token_usage`; the means are 13.9% and 25.2%), so realization is not a function of occupancy alone and a rule of the form "the pool collapses above $x$% occupancy" is false

on our own data. Occupancy is instantaneous; reuse depends on whether a prefix survives from one round to the next, which is a history the instantaneous number does not carry. We do not measure that history here, and we say so rather than fitting the two traces to a common curve.

*b) The elasticity criterion transfers no better here.:* On both traces the criterion plays the same role, and it is the modest one §IV settles on: it *rules out* the workload-side story. The controller on this trace runs at a busy-time mean of 1.223, which falls inside the span its own static points cover (1.077, 1.161, 1.287)—more precisely inside the cap3→cap4 segment, where $\varepsilon(\tau_1) = -0.28$ over $\overline{R}$ 1.16–1.29. The other measured segment, cap2→cap3, is $-2.40$. Both are negative, so nothing in this trace's measured range says added concurrency was self-defeating, and the controller's failure is again not explained by the criterion. What this trace cannot do is test the criterion's *expensive* branch: the highest running count it reaches—a mean of 1.287 over busy time, with an instantaneous peak of four in flight—never approaches the $\overline{R} \approx 1.5$ where the main trace's cliff begins. So the cliff is untested here, neither confirmed nor refuted, and we do not read this trace's negative result as evidence about it.

We also record what the noise does, because it is the same weakness the whole paper has. Recomputing each segment's $\varepsilon$ with $\tau_1$ perturbed by $\pm 10\%$, a perturbation inside the range our repeat pairs actually show (1.2–22.4%), gives $[-5.20, +0.41]$ for cap2→cap3 and $[-2.17, +1.61]$ for cap3→cap4. Only the second band crosses one. So the sign of $\varepsilon$ in the segment the controller actually occupies is not resolvable at that perturbation—which is a bound on the paragraph above, not a reprieve from it.

*c) The controller's verdict transfers, weakly.:* Foyer finishes the second trace in 75.6 minutes against 66.4 for the fastest static cap (on the three caps of `elasticity_x.csv` the fastest is 66.9), so it is 13.2% slower (13.15% at full precision)—but that gap is inside the run-to-run spread we measure elsewhere (up to 22.4% between repeat pairs of the same configuration), so we do not claim it lost. What we can say is that it did not win, on either axis: its signed distance from the second trace's static envelope is $+0.40$ pp on the SLO axis and $-0.14$ pp on the cache-hit axis, both inside the tie band. The SLO number is nearly uninformative, and we flag that rather than let it stand as a measurement: at a 10 s threshold every point on this trace scores 99.5–100.0%, so the axis cannot discriminate: the whole spread between its fastest and slowest configuration is half a point, inside the tie band itself.

### B. What this section does and does not buy

It buys a negative result that is not a property of one trace family: on a second workload, with a different session-size distribution and a tool-wait distribution about 60× longer at the median, the same controller still does not beat the same kind of static cap, and the reuse degradation still appears. It costs us the transportability of the mechanism. On the main trace the criterion narrowed the search by excluding the workload-side explanation; on this one it cannot even do that, because the controller never reaches the region the criterion describes. We report both, and we note that the second workload is closer to the regime where adaptive control *should* pay—lower concurrency, much longer idle gaps—and the controller still did not win there.

TABLE II
THE 25-SESSION TIER, ALL FOUR ARMS UNDER THE CURRENT HARNESS. THIS IS THE SAME-GENERATION COMPARISON EARLIER DRAFTS OF THIS PAPER WERE MISSING; `cap4` AND THE FOYER ARM WERE RUN FOR THIS REVISION.

| Policy | Wall (min) | Conc. | SLO (%) | Seg. SD (pp) |
|---|---|---|---|---|
| cap2 | 32.4 | 1.95 | 92.8 | 11.0 |
| cap3 | 24.2 | 2.77 | 91.2 | 5.1 |
| cap4 | 23.6 | 3.39 | 79.2 | 21.4 |
| Foyer | 27.6 | 2.47 | 92.8 | 7.1 |

## VII. LOAD DEPENDENCE

A negative result about adaptivity invites the obvious objection: adaptivity pays when conditions *change*, and a single stationary trace is the one case where a static cap should be near-optimal. We agree. What we test here is a weaker thing—two separate constant-load tiers—and §IX says what that leaves open.

We run the same controller and the same cap family on a 25-session tier. The throttle configuration is reversed: with all safety throttles disabled, the 25-session tier runs *51% faster* (58.4 to 28.8 minutes) at a cost of only 2.4 pp of SLO, while on the 200-session tier disabling the same throttles costs 42% (291.3 to 415.0 minutes) and 43.7 pp of SLO. One switch, two regimes, opposite signs, both around 40–50% (Fig. 7). No single scalar setting is optimal across both.

We flag the generation gap rather than let the figure imply more than we ran. The two 200-session numbers are current; the two 25-session numbers are from an earlier harness revision, and we never ran the all-throttles-off arm on the 25-session tier under the current one. So the reversal is a direction we measured across two harness generations, not a same-generation paired comparison, and a reviewer should read it that way. The same-generation 25-session measurement we do have is in Table II, and it does not contain that arm.

This is the regime in which adaptive control should win, and we now have the same-generation measurement that earlier drafts of this paper were missing. Four arms on the 25-session tier, all under the current harness (Table II): static caps of two, three and four finish in 32.4, 24.2 and 23.6 minutes at 92.8%, 91.2% and 79.2% SLO, and Foyer finishes in 27.6 minutes at 92.8%.

The finding is a trade, not a defeat, and it is the opposite of what the 200-session tier shows. Against the same-generation envelope—which runs from `cap4` at (23.6, 79.2) through `cap3` at (24.2, 91.2) to `cap2` at (32.4, 92.8)—Foyer sits $+0.9$ pp above the line. A static cap of three is still 12% faster in wall clock, but it gives up 1.6 pp of SLO to do it, so neither

dominates the other. On this tier the controller is a different point on the front rather than a dominated one, which is exactly what the 200-session tier denies it.

We are careful about the size, and about the yardstick. $+0.9$ pp is measured against a straight-line interpolation between static points, not against a switching policy anyone ran, so the defensible statement is that Foyer is not dominated by any static point we measured on this tier—not that it beats what a static mix could achieve. On 125 requests one request is worth 0.8 pp, so the margin is about one request. The tier's segment-to-segment SD (5.1 to 21.4 pp) is not the right yardstick: segment volatility is not run-to-run uncertainty. The right one is the replicate spread we have—which, across the five configurations we repeated, is 0.5, 0.8, 1.0, 3.9 and 5.1 pp (`replicates.csv`). Against the two tightest of those $+0.9$ pp is about a replicate wide; against the three widest—which belong to the configurations that decide our headline elsewhere—it is inside one. We quote the tightest pair because it is the least favourable to us, not because it is the whole distribution. What we will defend is the weaker claim: on the low-load tier the controller is not worse, and the same switch that is worth $+51\%$ there costs $-42\%$ at high load. A controller that is competitive where load is light and dominated where load is heavy is the pattern this paper set out to explain, and it is the pattern we measured.

The second finding is the more useful one. The signal an adaptive controller would need is available—the number of sessions waiting for admission differs by 6–7$\times$ between the tiers (median 9–11 at 25 sessions against 66 at 200, maximum 135)—but its *interpretation* is not obvious. At the 25-session tier the controller holds eleven sessions back while running exactly one; that backlog is manufactured by the throttling itself rather than evidence that demand has slackened. Reading it correctly requires combining instantaneous backlog with progress through the run, which is a stateful computation a cap cannot perform. We regard this as the most promising direction we did not finish, and we are explicit that we did not finish it.

## VIII. Related Work

*a) Admission and scheduling for agentic serving.:* Recent systems report large gains from treating an agent session, rather than a request, as the schedulable unit, and from preserving session state across tool calls: session-aware admission with CPU-restorable KV [5], [6], program-level pause/resume with shortest-context-first restoration [7], and congestion-based control over an agent-level window [8]. A recurring measurement in this line is that recomputation, not admission, dominates: [8] attributes 49.1% of end-to-end latency to recomputation, and [9] preserves state on the CPU at the cost of a reload. Our results are consistent with this literature rather than in tension with it, and §IX explains why: the mechanism these systems add is one our controller already has.

*b) Engine-side techniques.:* A separate line improves the engine rather than the admission decision. Iteration-level batching and paged KV [4], [10] are the substrate every policy here runs on; chunked prefill with stall-free batching [11], prefill/decode disaggregation [12], and cache-centric scheduling [13] are the refinements. Our §III measures 2.6–8.2 ms of each 27–37 ms step as work that is not decoding—prefill and non-decode overhead—which is exactly what this line attacks. We expect these techniques to move the frontier rather than to decide who sits above it, but we did not run the interaction and so state it as an expectation.

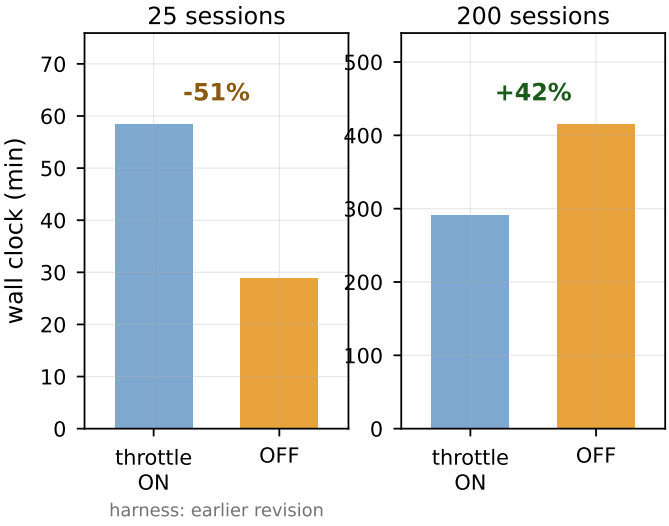



Fig. 7. The same throttle switch, two load tiers, opposite signs. The 25-session bars are from an earlier harness revision and are shown as a direction, not a magnitude—the current-harness 25-session comparison is in Table II instead. The 200-session bars are current.

*c) State retention and preemption.:* A third line keeps session state alive across the gaps we measure in §III: keeping KV on cheaper storage across turns [14], [15], migrating it between instances [16], and preempting at iteration granularity [17]. These systems exercise finer control than our 2 s admission cycle, which is the boundary of our negative result: we falsify one control granularity, not the family. What they share with Foyer is that the state is not discarded—which is why, as we argue below, adding admission intelligence on top buys nothing here.

*d) Application-level dependencies and replay.:* Serving an LLM *application* rather than a request means the scheduler sees dependencies between calls [18], and agentic traces have structure that request-level benchmarks do not capture [19]. Our replay preserves that structure but fixes it in advance, which is exactly the limitation these works would press on.

*e) Prefix caching.:* The prefix cache is often proposed as the signal an admission policy should use. What we measure is that the *realized* reuse is a function of the admission policy and of the host tier, while the *planned* reuse is a constant of the trace; an earlier draft of this paper conflated the two and got the direction wrong in both. The trace plans a round-1-and-later hit rate of 80.9%. Without HiCache the realized round-1+ rate follows the plan only at the loosest cap and then collapses: 80.8%, 78.5%, 70.6%, 56.0%, 32.8%

for caps of one through five. With HiCache it tracks the plan at every cap we ran: 80.7%, 79.6%, 78.7% for caps of two, three and four. Our controller sits at 80.6%. So the honest statement is not that admission does not move this quantity—`cap4` alone moves it 25 points—but that the host tier is what keeps it from moving, which is the same mechanism §III and §IV keep running into. A configuration without a host tier that admits four sessions does not have the cache that a host-tier one at the same concurrency does. All eight numbers in this paragraph are the *observed* per-step hit rate (`server_prefix_hit_rate` in the per-step logs), unweighted mean over rounds $\geq 1$; the planned rate is the trace constant 80.9% and does not move with the cap. §III uses a different convention—all steps, token-weighted, so that reuse and recomputation partition the prompt—under which the same collapse reads 92.6%→36.3% of the trace's bound for the device-only caps and 92.3%→90.2% with a host tier. We state both because the paper's earlier drafts mixed them. We note the boundary of this claim—we varied admission policy, not eviction policy, and SGLang exposes alternate eviction policies we did not test. A characterization of production KV-cache traffic at a large cloud provider finds that eviction-policy design is "highly workload-dependent" and that reuse patterns differ sharply by request category [20]; that is reason to expect a single trace family to understate the space, though it is not a policy comparison and we do not present it as one.

## IX. Discussion

*a) Why published gains do not reappear.:* The systems cited above win by keeping session state alive across tool waits and by pausing at agent granularity rather than request granularity. Foyer has both properties by construction: the permit file gives per-session pause and resume, and the controller never frees capacity by discarding a session—it admits fewer sessions instead. To that extent the mechanism these systems add is one our controller already has, which is one reason adding admission intelligence on top of it buys nothing here.

We are careful not to overstate this, because the distinction is between *choosing* to discard state and *having* it evicted. Retention here is the engine's radix cache doing its ordinary job; it is not something the controller implements, and §III measures that the cache evicts parked prefixes under pressure and that they are recomputed. What Foyer avoids is the deliberate trade of state for capacity. A system that manages the host tier as an explicit policy—as [9] and [14] do—holds a lever we do not, and our negative result does not cover it.

*b) The threshold decides the verdict, and we report that.:* Every SLO number in this paper counts a request as good if its time-to-first-token is under 10 s. That constant sets the whole vertical axis, so we re-derived the frontier at 5 s and 20 s as well, from the same per-request logs. The ordering of the static caps does not change. What changes is where the controllers sit relative to them, and at the loose end one of them crosses:

| SLO threshold | 5 s | 10 s | 20 s |
|---|---|---|---|
| `horizon`=1 | −2.9 | −0.9 | +1.5 |
| Foyer base | −2.3 | −1.3 | +0.6 |
| `bres` | −11.9 | −7.8 | −4.3 |
| Concur-style AIMD | −14.2 | −11.2 | −6.5 |

Two readings, and we give both. The first is that our headline claim is conditional: at the strict threshold that interactive serving actually cares about, no controller reaches the static front; at a loose one, one does. The second is that the crossing is still not a win, though the reason is more awkward than we first wrote. +1.5 pp at 20 s is larger than the replicate SLO spread of the three tightest pairs we have—0.5 pp for the base controller, 0.8 pp for `cap3`+HiCache and 1.0 pp for `cap2`+HiCache—but smaller than the two widest, 3.9 pp for the best arm's own repeat and 5.1 pp for `cap4`+HiCache, and those two are anchors. It is the anchors, not the controllers, that set the line the margin is measured against, so the comparison that matters is the one against the widest anchor—where the margin is not resolvable. (An earlier revision discarded the `cap3`+HiCache pair on the ground that one run had exceeded its cap. That was wrong: the over-cap run is the *device-only* `cap3` repeat, and both HiCache runs peak at three with none over. The pair is a valid replicate and we use it, including as an envelope anchor.) A margin that appears only at one threshold and only at a size we cannot resolve is not evidence that adaptive admission works; it is evidence that the question needs more runs than we have. We state the conditional version in the abstract, and we flag this as the second thing we would re-run with more GPU time.

*c) The SLO metric excludes the controller's own cost, and we checked.:* Every SLO number above counts a request as good if its time-to-first-token is under 10 s, and the runner measures that from the moment the request is *submitted to the engine*. A session that waits for a permit has not been submitted yet, so the wait is invisible—and that wait is precisely the cost an admission controller imposes. We therefore recomputed the runs we have per-step logs for on a ready-to-first-token basis, where a round's clock starts when the session is actually able to run (its arrival for round 0, the previous round's completion plus its tool wait otherwise). That is nineteen of the twenty-five 200-session runs in Table I; absent are the no-control baseline, the two HiCache anchor repeats and three controller repeats, and we do not extend the claim to them.

The correction is large. Every configuration loses 2.9–19.3 pp of SLO, and the loss is systematic: tight caps queue the hardest, so `cap1` falls from 94.8% to 76.2% while `cap5` falls only from 27.1% to 24.2%. A session admitted last under `cap1` can wait hours for its turn; its engine-side TTFT is still 1.6 s.

What does *not* change is the verdict, within the subset we recomputed. The static envelope keeps the same four anchors, and of the eleven controller arms, nine move by no more than 2.7 pp and none crosses the $\pm 1.0$ pp band into a new verdict. Two move more: the all-throttles-off arm improves by 8.3 pp

and the retuned AIMD arm by 2.8 pp, both of which stay below the front either way. The one that decides our headline moves the wrong way for the controller: on the ready clock `horizon=1`'s own SLO falls from 84.8% to 68.2% while the anchor it is measured against falls less, so the ready clock moves it further below the line rather than closer to it. We do not give a ready-clock distance for it, because the two anchor repeats that define the line are not in the set we could recompute (§IX says which runs are), and inventing a line for them would be the same error this subsection is about.

The subset matters and we state its edge. The two configurations whose repeats decide how we read the verdict—`horizon=1`'s repeat and the fastest anchor `cap4`+HC's—are not in the nineteen, because we did not recompute their per-step logs on the ready clock; the rebuilt envelope therefore rests on the original anchors without their repeats, and a ready-clock frontier that included them could move the line by the 3.9 and 5.1 pp those repeats differ by. We do not claim the verdict is unchanged under every SLO definition—only that on the runs we could recompute, moving the clock to the ready point costs every configuration 3–19 pp and changes no verdict in that set. We report this because a reviewer should ask it, and because the answer is the useful part: the negative result is not an artifact of choosing an SLO definition that flatters controllers. If anything the flattering choice was the one that moved our best arm's distance from $-0.9$ to $+1.5$—that is, from just below the line to just above the tie band's upper edge, by a margin of the same order as that pair's own replicate spread (3.9 pp).

*d) Limitations.:* Our results come from one 24 GB accelerator, one 7B model and one trace family; generality is untested. Only five configurations have been repeated—four of them twice and the base configuration four times—so the replicate spread we report in §V is estimated from one to three degrees of freedom per configuration; a margin as small as the one that decides our verdict would need several more repeats per configuration to resolve, and we did not run them. Two specific boundaries matter. First, the pool holds about four median sessions, so concurrency is capacity-bound by construction; on a device with a much larger pool the premise may not hold and the conclusion may invert—we regard reproducing on a larger device as the single most important next experiment. Second, the load is stationary, single-tenant and single-SLO. The load-dependence result of §VII is measured on two *separate* constant-load tiers, so it says what to expect when conditions differ, not what happens while they change. We did run the experiment that would settle that—a trace whose load moves mid-run, with a static cap on each tier and a switcher that changes its cap at the trace's designed switch point—and we cannot report a result from it, in either direction, because its static baselines were measured under host co-tenancy: `var_cap3` spent 158 of its 266 minutes at four or more concurrent engine instances, and `var_cap2` is flagged as a watch (§IX-0e). A first look at those runs has the switcher finishing 15 minutes ahead of the best static cap at an SLO inside the tie band, which would be the first evidence in this paper that adaptivity pays; we do not report it as a result, because the baseline it beats is precisely the one the audit flags, and a 60%-contaminated baseline is inflated in exactly the direction that manufactures that finding. Clean re-runs of that family are in flight, and the honest position until they land is that the variable-load case is measured and unusable, not unmeasured. Multi-tenant and heterogeneous-SLO settings are where we would look next.

*e) Reproducibility.:* The front and the signed distance of each controller are computed by `scripts/lab/frontier.py` from the raw per-step logs; the run tables and the four runs shorter than 999 steps are computed by `scripts/lab/gen_table.py` and `scripts/lab/run_summary.py` respectively, and §IV is computed by `gen_elasticity.py` and `gen_elasticity_hc.py`. Every run is guarded against a shrunken KV pool.

*Co-tenancy on the measurement host.* Our runs share a host with other tenants and with each other, and late in the campaign we found that this matters more than the pool guard catches. An audit (`gen_concurrency.py`, data in `paper/data/concurrency.csv`) reconstructs, for every run, how many other runs were alive for how long. Runs that overlapped three or more others for at least an hour are flagged; fourteen of ours are, two of which are failed runs we do not report. The flag is a risk marker, not a measurement of damage, and we treat it that way: three of the twelve (`F:t85`, `F:t90`, `var_foyer`) show no aggregate inflation at all, because their overlap was short relative to the run. Where the damage is legible it is large—the per-request decode interval $\tau_2$, measured client-side and never touching the sampler, sits at 30.9–33.3 ms for the four affected arms in Table I against 23.8–25.7 ms for the same family measured clean. Two controls pin the cause to co-tenancy rather than to the configuration. One run (`cap1`) holds $\tau_2$ at 20.3–20.7 ms for its first five and a half hours and jumps to 22.4–27.5 ms in the minute two other runs start. And two independent runs (`F:t85`, `F:rl99`) step down from ≈30 ms to 22–25 ms in the same wall-clock minute, when a third run ends. Fourteen of our runs fall in the flagged class; six of them are rows of Table I (`cap1`, `F:bres`, `F:hz0`, `F:hz2`, `F:breshz1`, and the published-parameter AIMD row) and eight are arms we do not report here. The flag is a window, and only four of those six are actually damaged—the four controller variants whose $\tau_2$ is 30.9–33.3 ms. `cap1` and the AIMD row show no aggregate inflation (20.6 and 24.4 ms), so we keep their numbers and flag the window. The four damaged rows are being re-measured under the same one-run-per-GPU discipline as everything else, and until those runs land we treat their wall-clock, SLO and step-cost columns as unusable: the signed distances in Table I for those four rows, the step-cost attribution of §V-C where it names them, and the route-1 comparison of §V which rests on `bres`, are all provisional. We flag them rather than quietly dropping the rows so that a reader can see which conclusions are still standing on which measurements.

Two things we can already say. First, the failure is silent:

the engine-side counters—running count, pool occupancy, evictions, recomputation—are indistinguishable between the affected runs and the clean ones, so nothing in the ordinary telemetry would have caught this. Second, the affected runs' *volume* columns look untouched: the four damaged host-tier arms reuse 0.6785–0.6950 of the prompt against 0.6347–0.6975 for the six clean host-tier arms, a range overlap that is suggestive but not proof—co-tenancy could in principle also reorder completions and change eviction order. The re-runs will settle it, since they are the same policies measured clean, and we compare reuse per round and per session between the pairs rather than only the totals.

One caveat about the controller artifact, which a reader with our code would otherwise hit. There is no single controller copy behind this paper: the runs record four different hashes of `controller_budget3.py` (`2b573829083c` on 31 runs, `f825bc768188` on 24, `d692ab11f396` on 4 and `e1dd5ae78019` on 3), so the artifact evolved during the campaign and a reader cannot reproduce every number from one revision. None of the four matches the repository copy (`a21bd2b63774`), and the difference is not one hunk but several: the queue-scan fix described in §V-H, and a rename of the logging variable whose rebinding §V identifies as the source of a retraction. The repository has both; none of the four running copies has either. The numbers in this paper therefore describe the pre-fix path, with one partial exception we want to state precisely rather than let a reader assume the stronger thing. We did run the same configuration with `sf=0` (§V-B), and on the running copy that changes two things at once: it scans the queue *and* drops the one-per-cycle limit, because the `break` that enforces single-flight is inside the same conditional. The repository's fix keeps the limit and only adds the scan. So our run is not a test of the fix as written—it is a test of the fix plus a looser admission rate, and it moves the arm along the front rather than past it. The narrower experiment, scan-with-the-limit-kept, is the one we would do first with more GPU time. We report six claims we retracted during the work, because the retractions were the most informative events. In order of how much they cost us:

*The denominator of the step cost.* Our first §III divided by the whole-run mean running count while multiplying by busy time. The two are not the same quantity—the whole-run mean includes the idle samples—so $\tau$ came out low by exactly the busy fraction, 36% for `cap1`, and two independent errors cancelled to make our two measurement methods appear to agree. This is the one that changed a headline number.

*A mechanism refutation that was itself refuted*: we withdrew the eviction-to-recomputation account on the strength of two checks that were both identities rather than measurements (§III).

*A median-based reading of the tool wait* that inverted a central assumption. *A monotone relation between idle fraction and concurrency* that the full data does not show. *A growth-sensitivity conclusion* that rested on an arm we had already invalidated. And *the accounting double-charge* of §V, falsified twice—once by source inspection and once by the end-to-end measurement here.

## X. Conclusion

We asked whether an adaptive admission controller can beat a tuned static concurrency cap on a memory-bound agentic LLM workload, and found that on our system it cannot: nine variants, four routes tested, none above the static front. We expected the reason to be a cancellation—admission policy controls how many requests are in flight, the per-step cost rises with that number, and the product is what the wall clock sees—and we built a two-part decomposition of the wall clock that makes the cancellation measurable. That decomposition holds, and the cost it measures is real, but it is not the whole reason and on the configuration our controllers run it is not the binding one: there the elasticity stays below one across the operating range, so added concurrency was still cheap and the engine had room. Our controllers did not take it. What we can say at the end is narrower than what we set out to say: on this hardware and this workload, the admission rules we could build and tune did not beat a static cap, the capacity estimate they act on is $1.79\times$ the truth, and a structural idle component that no policy we built recovers at a profit remains. The regime in which adaptivity should have paid—where the same switch is worth $+51\%$ and $-42\%$—is one where the controller is not dominated by any static point we measured, and does not dominate one either. The measurement we offer is a diagnostic an engineer can take before committing to a sweep; whether it predicts the outcome on another workload is a question we tested once and could not confirm.

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